Internet Traffic Monitoring and Analysis:

Wireshark Tutorial

Outline

What is Wireshark?

Capturing Packets

Analyzing Packets

Filtering Packets

Saving and Manipulating Packets

Packet Statistics

Colorizing Specific Packets

References

What is Wireshark?

The De-Facto Network Protocol Analyzer

- Open-Source (GNU Public License)
- Multi-platform (Windows, Linux, OS X, Solaris, FreeBSD, NetBSD, and others)
- Easily extensible
- Large development group

Previously Named “Ethereal”

What is Wireshark?

Features

- Deep inspection of thousands of protocols
- Live capture and offline analysis
- Standard three-pane packet browser
- Captured network data can be browsed via a GUI, or via the TTY-mode TShark utility
- The most powerful display filters in the industry
- Rich VoIP analysis
- Live data can be read from Ethernet, IEEE 802.11, PPP/HDLC, ATM, Bluetooth, USB, Token Ring, Frame Relay, FDDI, and others
- Coloring rules can be applied to the packet list for quick, intuitive analysis
- Output can be exported to XML, PostScript®, CSV, or plain text

What is Wireshark?

What we can:

- Capture network traffic
- Decode packet protocols using dissectors
- Define filters – capture and display
- Watch smart statistics
- Analyze problems
- Interactively browse that traffic

Some examples people use Wireshark for:

- Network administrators: **troubleshoot network problems**
- Network security engineers: **examine security problems**
- Developers: **debug protocol implementations**
- People: **learn network protocol internals**

Tools Overview

Tcpdump

- Unix-based command-line tool used to intercept packets
  - Including filtering to just the packets of interest
- Reads "live traffic" from interface specified using **`-i`** option …
- … or from a previously recorded *trace file* specified using **`-r`** option
  - You create these when capturing live traffic using **`-w`** option

Tshark

- Tcpdump-like capture program that comes w/ Wireshark
- Very similar behavior & flags to tcpdump

Wireshark

- GUI for displaying tcpdump/tshark packet traces

Tcpdump example

- Ran tcpdump on a Unix machine
- First few lines of the output:

```
01:46:28.808262 IP danjo.CS.Berkeley.EDU.ssh > adsl-69-228-230-
 7.dsl.pltn13.pacbell.net.2481: . 2513546054:2513547434(1380) ack
 1268355216 win 12816

01:46:28.808271 IP danjo.CS.Berkeley.EDU.ssh > adsl-69-228-230-
 7.dsl.pltn13.pacbell.net.2481: P 1380:2128(748) ack 1 win 12816

01:46:28.808276 IP danjo.CS.Berkeley.EDU.ssh > adsl-69-228-230-
 7.dsl.pltn13.pacbell.net.2481: . 2128:3508(1380) ack 1 win 12816

01:46:28.890021 IP adsl-69-228-230-7.dsl.pltn13.pacbell.net.2481 >
 danjo.CS.Berkeley.EDU.ssh: P 1:49(48) ack 1380 win 16560
```

What does a line convey?

Timestamp

This is an IP packet

Source port number (22)

01:46:28.808262 IP danjo.CS.Berkeley.EDU.ssh >

adsl-69-228-230-7.dsl.pltn13.pacbell.net.2481: .

2513546054:2513547434(1380) ack 1268355216 win 12816

Destination host name

Destination port number

TCP specific information

- Different output formats for different packet types

Similar Output from Tshark

1190003744.940437 61.184.241.230 -> 128.32.48.169 SSH Encrypted request packet len=48

1190003744.940916 128.32.48.169 -> 61.184.241.230 SSH Encrypted response packet len=48

1190003744.955764 61.184.241.230 -> 128.32.48.169 TCP 6943 > ssh [ACK] Seq=48 Ack=48 Win=65514 Len=0 TSV=445871583 TSER=632535493

1190003745.035678 61.184.241.230 -> 128.32.48.169 SSH Encrypted request packet len=48

1190003745.036004 128.32.48.169 -> 61.184.241.230 SSH Encrypted response packet len=48

1190003745.050970 61.184.241.230 -> 128.32.48.169 TCP 6943 > ssh [ACK] Seq=96 Ack=96 Win=65514 Len=0 TSV=445871583 TSER=632535502

Demo 1 – Basic Run

Syntax:

tcpdump [options] [filter expression]

Unfortunately, Eustis machine does not allow normal users to run tcpdump

I will demonstrate it on my group's Unix machine: acnserver.fils.edu

- $ sudo tcpdump –i eth0
- On your own Unix machine, you can run it using "sudo" or directly run "tcpdump"

Observe the output

Filters

We are often not interested in all packets flowing through the network

Use filters to capture only packets of interest to us

Demo 2

1. Capture only udp packets
   - tcpdump “udp”
2. Capture only tcp packets
   - tcpdump “tcp”

Demo 2 (contd.)

1. Capture only UDP packets with destination port 53 (DNS requests)
   - tcpdump “udp dst port 53”
2. Capture only UDP packets with source port 53 (DNS replies)
   - tcpdump “udp src port 53”
3. Capture only UDP packets with source or destination port 53 (DNS requests and replies)
   - tcpdump “udp port 53”

Demo 2 (contd.)

1. Capture only packets destined to quasar.cs.berkeley.edu
   - tcpdump “dst host quasar.cs.berkeley.edu”
2. Capture both DNS packets and TCP packets to/from quasar.cs.berkeley.edu
   - tcpdump “(tcp and host quasar.cs.berkeley.edu) or udp port 53”

How to write filters

Refer the tcpdump/tshark man page

Many example webpages on the Internet

Running tcpdump

Requires superuser/administrator privileges on Unix

- http://www.tcpdump.org/
- You can do it on your own Unix machine
- You can install a Linux OS in Vmware on your machine

Tcpdump for Windows

- WinDump: http://www.winpcap.org/windump/
  - Free software

So What is WireShark?

Packet sniffer/protocol analyzer

Open Source Network Tool

Latest version of the ethereal tool

Wireshark Interface

Tucker Ellis & West aaa.pcap - Wireshark

File Edit View Go Capture Analyze Statistics Help

Filter: Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
1	0.000000	192.168.1.2	192.168.1.255	NBNS	Name query NB ECI_DOMAIN<1c>
2	0.746308	192.168.1.2	192.168.1.255	NBNS	Name query NB ECI_DOMAIN<1c>
3	0.751270	192.168.1.2	192.168.1.255	NBNS	Name query NB ECI_DOMAIN<1c>
4	9.318731	Silicom_01:6e:bd	Broadcast	ARP	Who has 192.168.1.1? Tell 19
5	0.000664	Castlene_00:34:56	Silicom_01:6e:bd	ARP	192.168.1.1 is at 00:30:54:00
6	0.000026	192.168.1.2	192.168.1.1	DNS	Standard query A sip.cybercit
7	0.995383	192.168.1.2	192.168.1.1	DNS	Standard query A sip.cybercit
8	2.003039	192.168.1.2	192.168.1.1	DNS	Standard query A sip.cybercit
9	0.169652	192.168.1.1	192.168.1.2	DNS	Standard query response A 212
10	1.006246	192.168.1.2	192.168.1.1	DNS	Standard query SRV _sip._udp.
11	0.996899	192.168.1.2	192.168.1.1	DNS	Standard query SRV _sip._udp.
12	2.003024	192.168.1.2	192.168.1.1	DNS	Standard query SRV _sip._udp.
13	0.992343	Castlene_00:34:56	Silicom_01:6e:bd	ARP	Who has 192.168.1.2? Tell 19
14	0.000049	Silicom_01:6e:bd	Castlene_00:34:56	ARP	192.168.1.2 is at 00:e0:ed:01
15	1.010378	192.168.1.2	192.168.1.1	DNS	Standard query SRV _sip._udp.
16	4.005777	192.168.1.2	192.168.1.1	DNS	Standard query SRV _sip._udp.
17	8.002019	192.168.1.2	192.168.1.1	DNS	Standard query PTR 1.0.0.127.
18	0.001489	192.168.1.1	192.168.1.2	DNS	Standard query response PTR l
19	0.001640	192.168.1.2	212.242.33.35	SIP	Request: REGISTER sip:sip.cyb

```
    Header length: 20 bytes
  ⊞ Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00)
    Total Length: 78
    Identification: 0x698c (27020)
  ⊞ Flags: 0x00
    Fragment offset: 0
```

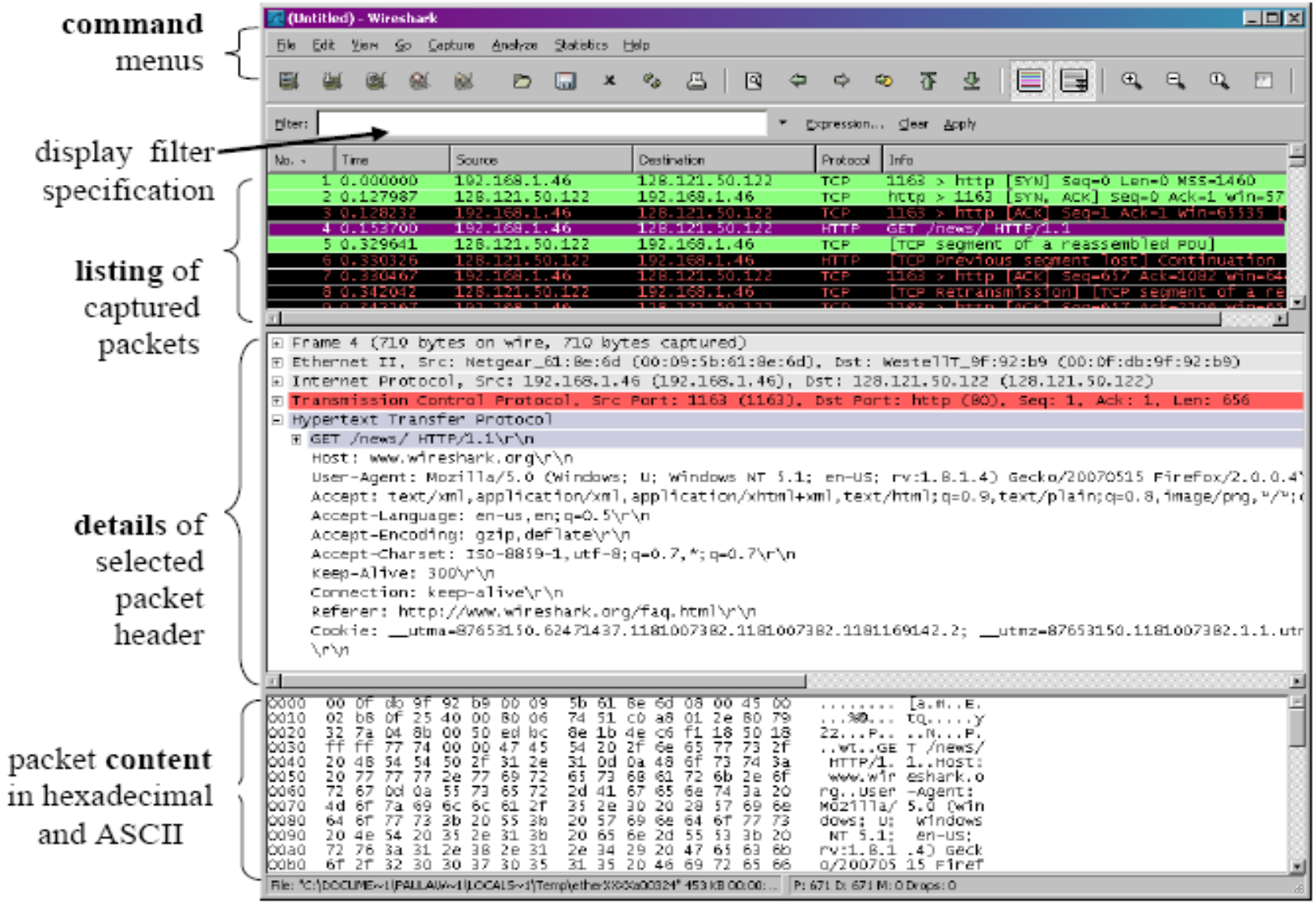
Highlighted packets here

```
0000  ff ff ff ff ff ff 00 e0  ed 01 6e bd 08 00 45 00   ........ ..n...E.
0010  00 4e 69 8c 00 00 80 11  4c c1 c0 a8 01 02 c0 a8   .Ni..... L.......
0020  01 ff 00 89 00 89 00 3a  5b b4 84 e7 01 10 00 01   .......: [.......
0030  00 00 00 00 00 00 20 45  46 45 44 45 4a 46 50 45   ...... E FEDEJFPE
0040  45 45 50 45 4e 45 42 45  4a 45 4f 43 41 43 41 43   EEPENEBE JEOCACAC
0050  41 43 41 43 41 42 4d 00  00 20 00 01               ACACABM. . ..
```

Are shown here as well

Identification (ip.id), 2 bytes | Packets: 691 Displayed: 691 Marked: 0 | Profile: Default

Wireshark Interface



Intel(R) PRO/Wireless 3945ABG Network Connection (Microsoft's Packet Scheduler) : Capturing - Wireshark

File Edit View Go Capture Analyze Statistics Telephony Tools Help

Filter: Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
64	36.858576	192.168.2.100	10.100.102.2	ICMP	Echo (ping) request
65	36.863613	10.100.102.2	192.168.2.100	ICMP	E
66	44.406189	192.168.2.100	10.100.102.1	SNMP	g ::ifOperS
67	44.413024	10.100.102.1	192.168.2.100	SNMP	g B::ifOper
68	44.499055	Msi_d4:52:4d	Broadcast	ARP	W 1? Tell
69	45.609033	192.168.2.100	10.40.41.2	ICMP	E t
70	47.797985	192.168.2.100	10.40.41.2	ICMP	E t
71	48.891533	192.168.2.100	10.100.102.1	SNMP	get-request IF-MIB::ifOperS
72	48.897871	10.100.102.1	192.168.2.100	SNMP	get-response IF-MIB::ifOper
73	49.989403	192.168.2.100	10.40.41.2	ICMP	Echo (ping) request
74	53.048866	192.168.2.100	255.255.255.255	UDP	Source port: 1027 Destinat

Packet List

```
⊞ Frame 32 (86 bytes on wire, 86 bytes captured)
⊞ Ethernet II, Src: IntelCor_a2:d8:9a (00:1c:bf:a2:d8:9a), Dst: EdimaxTe_6e:2f:7d (00:0e:2e:6e:2f:7d)
⊞ Internet Protocol, Src: 192.168.2.100 (192.168.2.100), Dst: 10.100.102.1 (10.100.102.1)
⊟ User Datagram Protocol, Src Port: solid-mux (1029), Dst Port: snmp (161)
    Source port: solid-mux (1029)
    Destination port: snmp (161)
    Length: 52
  ⊞ Checksum: 0xa175 [validation disabled]
⊟ Simple Network Management Protocol
    version: version-1 (0)
```

Packet Details

```
0000  00 0e 2e 6e 2f 7d 00 1c  bf a2 d8 9a 08 00 45 00   ...n/}.. ......E.
0010  00 48 04 d4 00 00 80 11  02 60 c0 a8 02 64 0a 64   .H...... .`...d.d
0020  66 01 04 05 00 a1 00 34  a1 75 30 2a 02 01 00 04   f......4 .u0*....
0030  06 70 75 62 6c 69 63 a0  1d 02 03 00 e2 af 02 01   .public. ........
0040  00 02 01 00 30 10 30 0e  06 0a 2b 06 01 02 01 02   ....0.0. ..+.....
0050  02 01 08 05 05 00                                  ......
```

Packet Bytes

Frame (frame), 86 bytes | Packets: 74 Displayed: 74 Marked: 0 | Profile: Default

Status Bar

Tucker Ellis & West aaa.pcap - Wireshark

File Edit View Go Capture Analyze Statistics Help

Filter: Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
1	0.000000	192.168.1.2	192.168.1.255	NBNS	Name query NB ECI_DOMAIN<1c>
2	0.746308	192.168.1.2	192.168.1.255	NBNS	Name query NB ECI_DOMAIN<1c>
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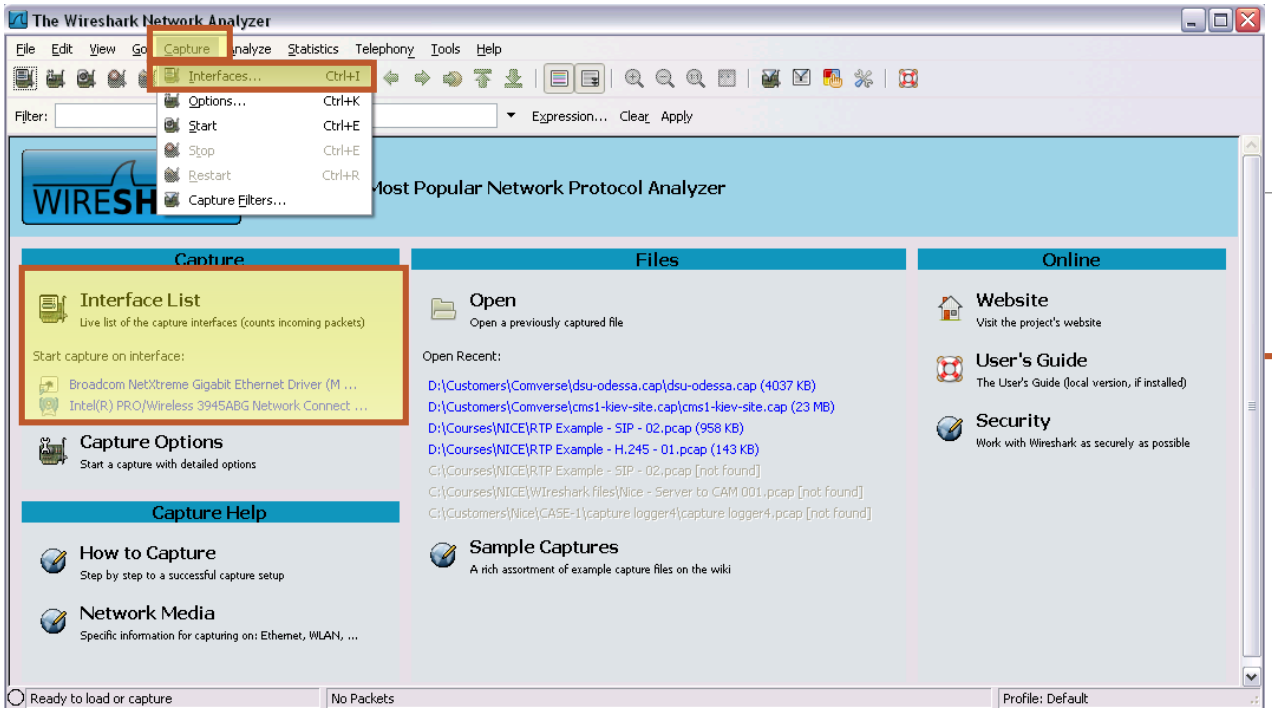
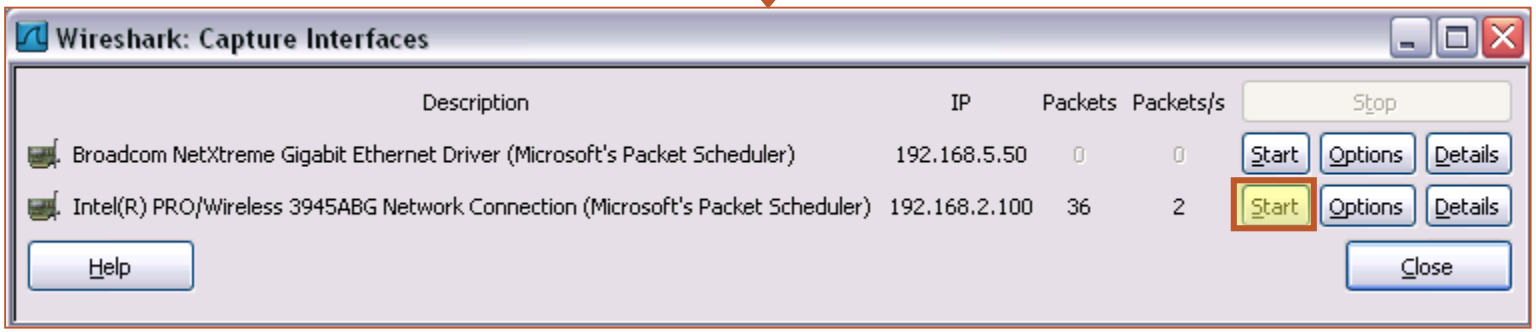
4	9.318731	Silicom_01:6e:bd	Broadcast	ARP	Who has 192.168.1.1? Tell 19
5	0.000664	Castlene_00:34:56	Silicom_01:6e:bd	ARP	192.168.1.1 is at 00:30:54:00
6	0.000026	192.168.1.2	192.168.1.1	DNS	Standard query A sip.cybercit
7	0.995383	192.168.1.2	192.168.1.1	DNS	Standard query A sip.cybercit
8	2.003039	192.168.1.2	192.168.1.1	DNS	Standard query A sip.cybercit
9	0.169652	192.168.1.1	192.168.1.2	DNS	Standard query response A 212
10	1.006246	192.168.1.2	192.168.1.1	DNS	Standard query SRV _sip._udp.
11	0.996899	192.168.1.2	192.168.1.1	DNS	Standard query SRV _sip._udp.
12	2.003024	192.168.1.2	192.168.1.1	DNS	Standard query SRV _sip._udp.
13	0.992343	Castlene_00:34:56	Silicom_01:6e:bd	ARP	Who has 192.168.1.2? Tell 19
14	0.000049	Silicom_01:6e:bd	Castlene_00:34:56	ARP	192.168.1.2 is at 00:e0:ed:01
15	1.010378	192.168.1.2	192.168.1.1	DNS	Standard query SRV _sip._udp.
16	4.005777	192.168.1.2	192.168.1.1	DNS	Standard query SRV _sip._udp.
17	8.002019	192.168.1.2	192.168.1.1	DNS	Standard query PTR 1.0.0.127.
18	0.001489	192.168.1.1	192.168.1.2	DNS	Standard query response PTR l
19	0.001640	192.168.1.2	212.242.33.35	SIP	Request: REGISTER sip:sip.cyb

```
    Header length: 20 bytes
  ⊞ Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00)
    Total Length: 78
    Identification: 0x698c (27020)
  ⊞ Flags: 0x00
    Fragment offset: 0
```

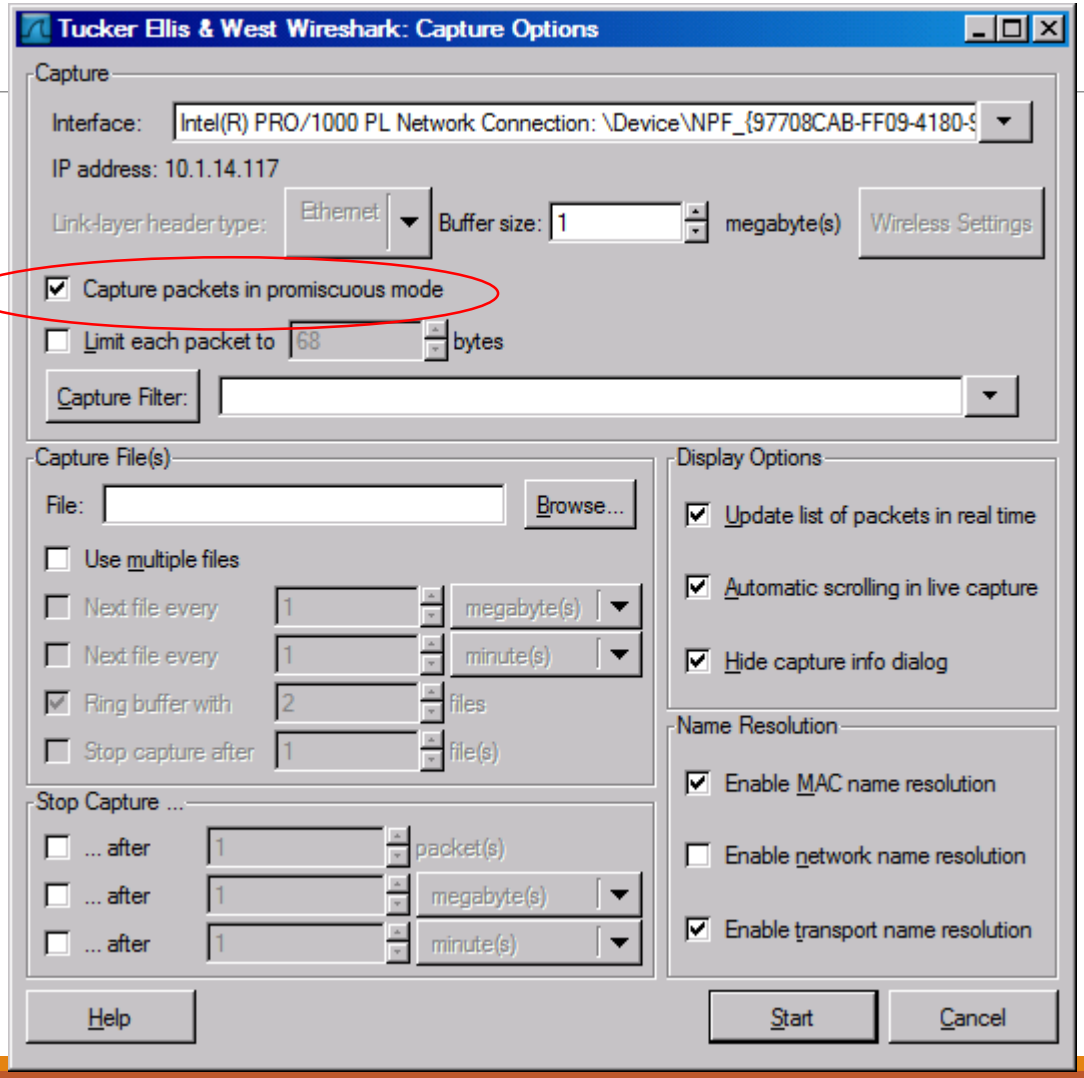
```
0000  ff ff ff ff ff ff 00 e0  ed 01 6e bd 08 00 45 00   ........ ..n...E.
0010  00 4e 69 8c 00 00 80 11  4c c1 c0 a8 01 02 c0 a8   .Ni..... L.......
0020  01 ff 00 89 00 89 00 3a  5b b4 84 e7 01 10 00 01   .......: [.......
0030  00 00 00 00 00 00 20 45  46 45 44 45 4a 46 50 45   ...... E FEDEJFPE
0040  45 45 50 45 4e 45 42 45  4a 45 4f 43 41 43 41 43   EEPENEBE JEOCACAC
0050  41 43 41 43 41 42 4d 00  00 20 00 01               ACACABM. . ..
```

Identification (ip.id), 2 bytes | Packets: 691 Displayed: 691 Marked: 0 | Profile: Default

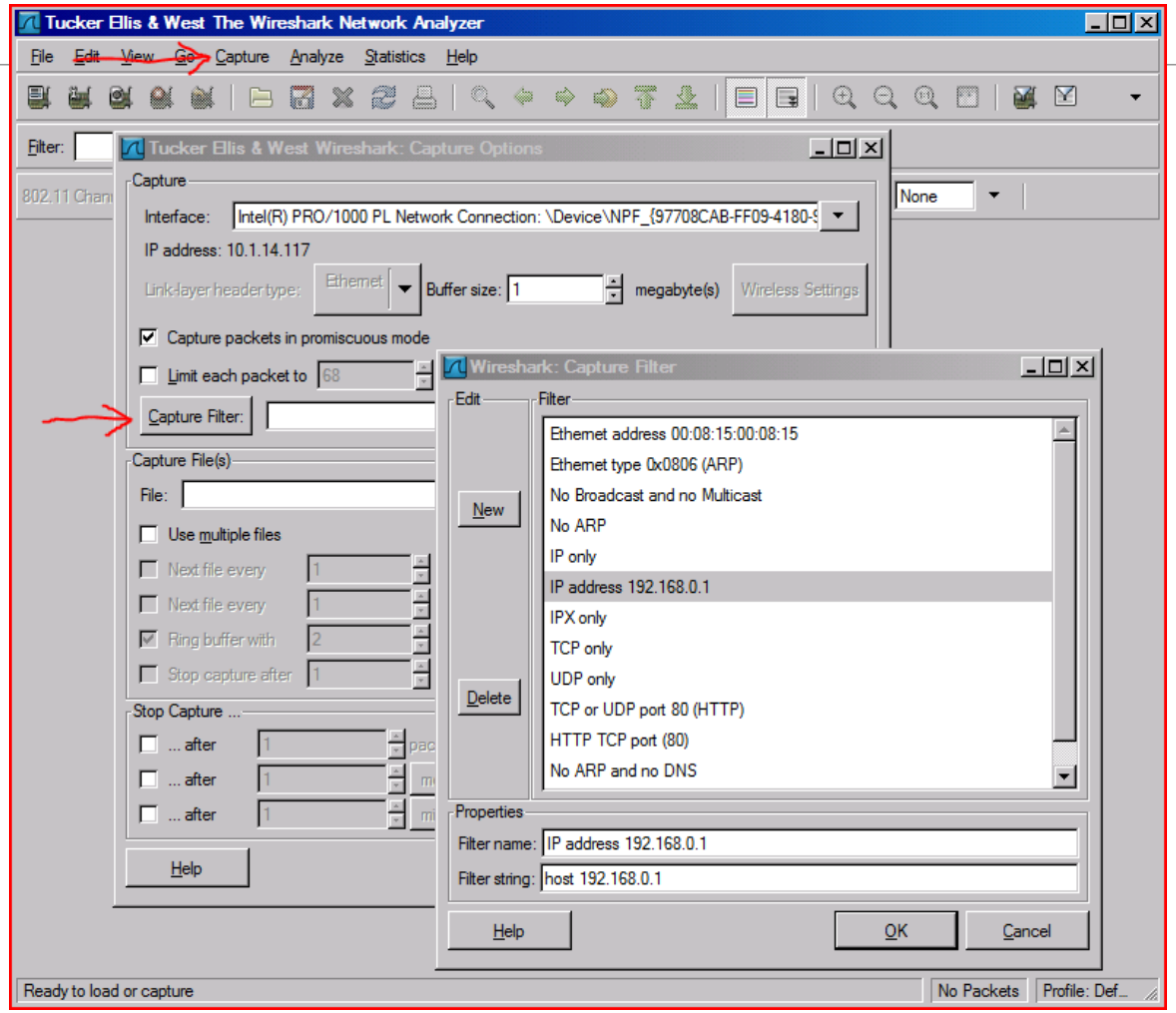

The Wireshark Network Analyzer
File Edit View Go Capture Analyze Statistics Telephony Tools Help
Interfaces... Ctrl+I
Options... Ctrl+K
Start Ctrl+E
Stop Ctrl+E
Restart Ctrl+R
Capture Filters...
Filter:
Expression... Clear Apply
Most Popular Network Protocol Analyzer
Capture
Interface List
Live list of the capture interfaces (counts incoming packets)
Start capture on interface:
Broadcom NetXtreme Gigabit Ethernet Driver (M ...
Intel(R) PRO/Wireless 3945ABG Network Connect ...
Capture Options
Start a capture with detailed options
Capture Help
How to Capture
Step by step to a successful capture setup
Network Media
Specific information for capturing on: Ethernet, WLAN, ...
Files
Open
Open a previously captured file
Open Recent:
D:\Customers\Comverse\dsu-odessa.cap\dsu-odessa.cap (4037 KB)
D:\Customers\Comverse\cms1-kiev-site.cap\cms1-kiev-site.cap (23 MB)
D:\Courses\NICE\RTP Example - SIP - 02.pcap (958 KB)
D:\Courses\NICE\RTP Example - H.245 - 01.pcap (143 KB)
C:\Courses\NICE\RTP Example - SIP - 02.pcap [not found]
C:\Courses\NICE\Wireshark files\Nice - Server to CAM 001.pcap [not found]
C:\Customers\Nice\CASE-1\capture logger4\capture logger4.pcap [not found]
Sample Captures
A rich assortment of example capture files on the wiki
Online
Website
Visit the project's website
User's Guide
The User's Guide (local version, if installed)
Security
Work with Wireshark as securely as possible
Ready to load or capture
No Packets
Profile: Default


Wireshark: Capture Interfaces
Description
IP
Packets
Packets/s
Stop
Broadcom NetXtreme Gigabit Ethernet Driver (Microsoft's Packet Scheduler)
192.168.5.50
0
0
Start
Options
Details
Intel(R) PRO/Wireless 3945ABG Network Connection (Microsoft's Packet Scheduler)
192.168.2.100
36
2
Start
Options
Details
Help
Close

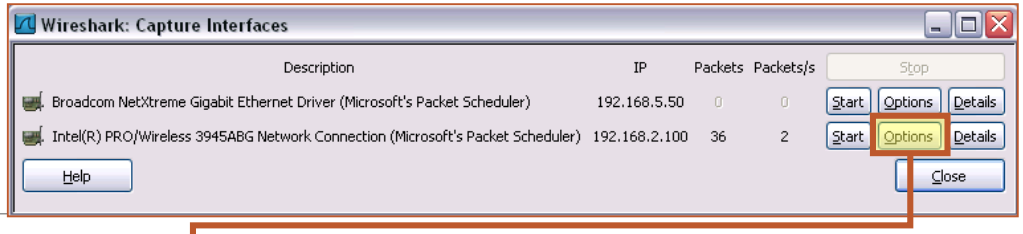
Capture Options



Capture Filter

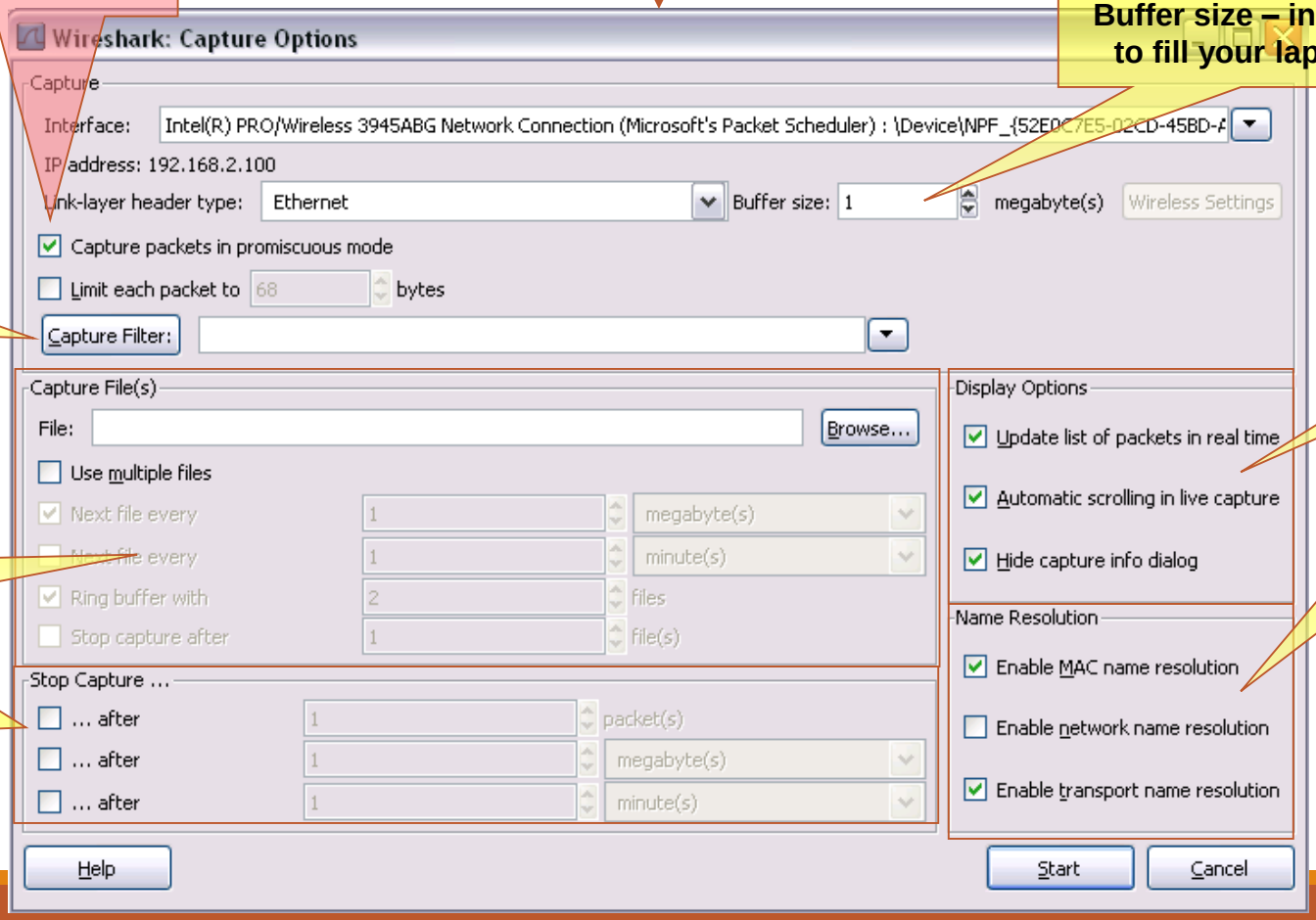


Capturing Packets (2/3)



Capture all packets on the network

Buffer size – in order not to fill your laptop disk



Capture filter

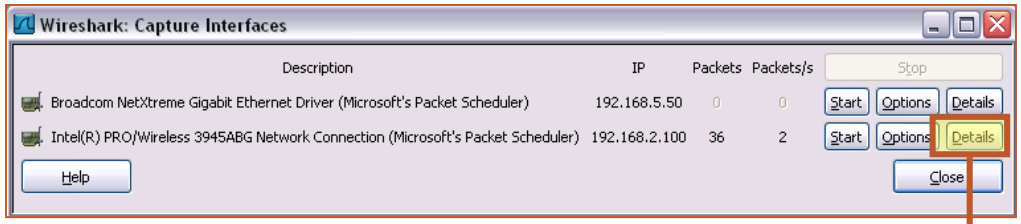
Display options

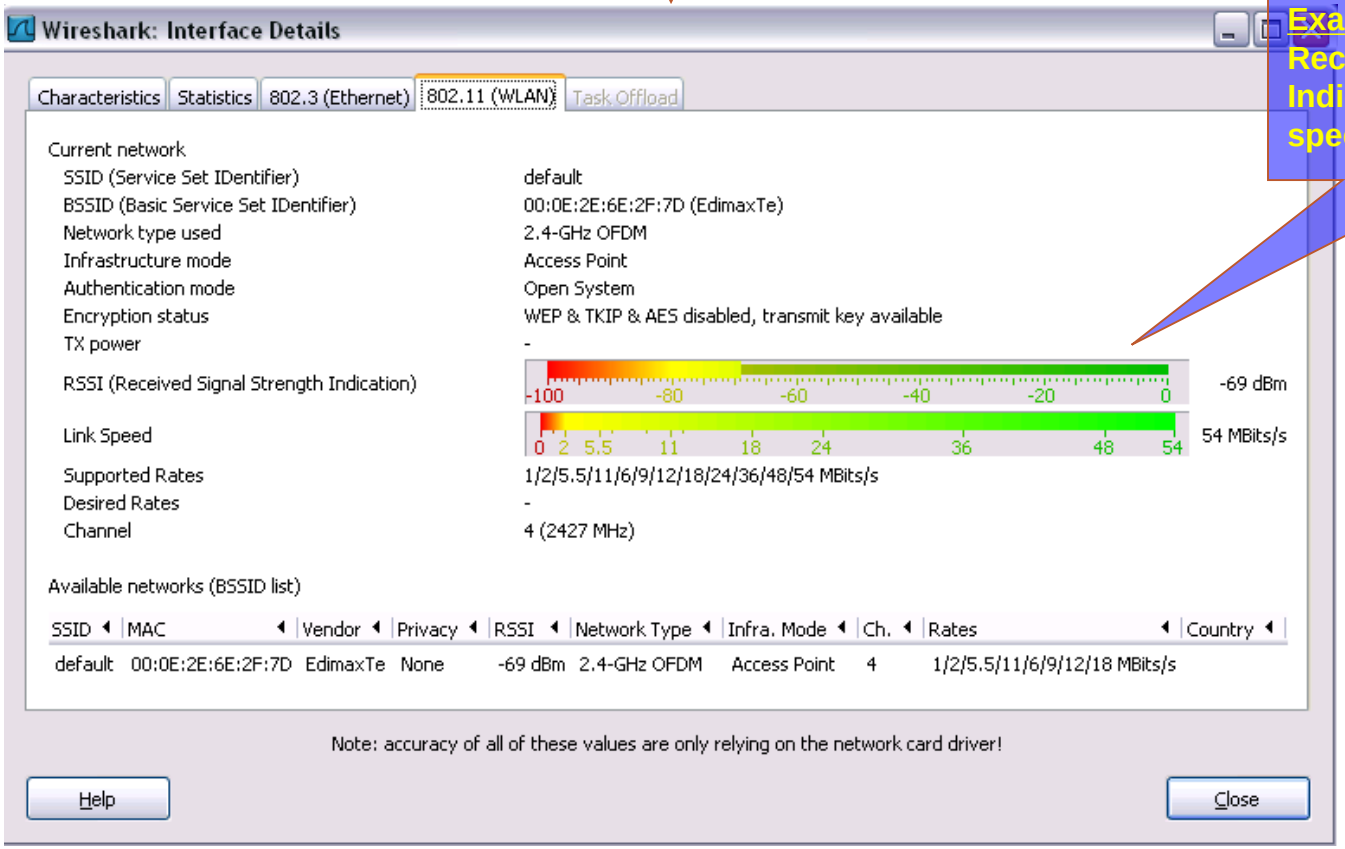
Capture in multiple files

Name resolution options

When to automatically stop the capture

Capturing Packets (3/3)





Example (W-LAN):
Received Signal Strength Indication (RSSI) and Link speed (BW)

Capture Filter examples

host 10.1.11.24

host 192.168.0.1 and host 10.1.11.1

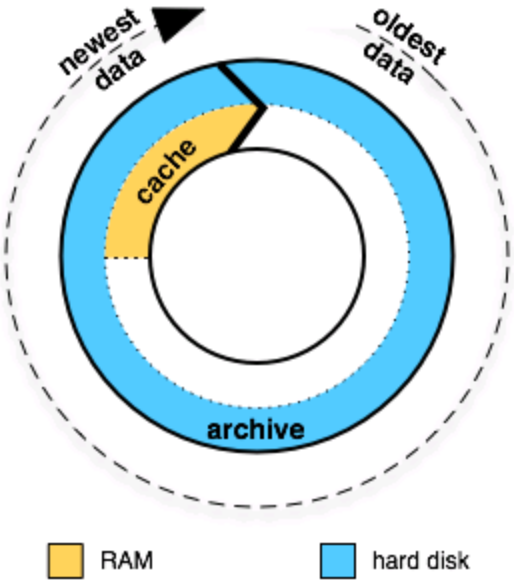
tcp port http

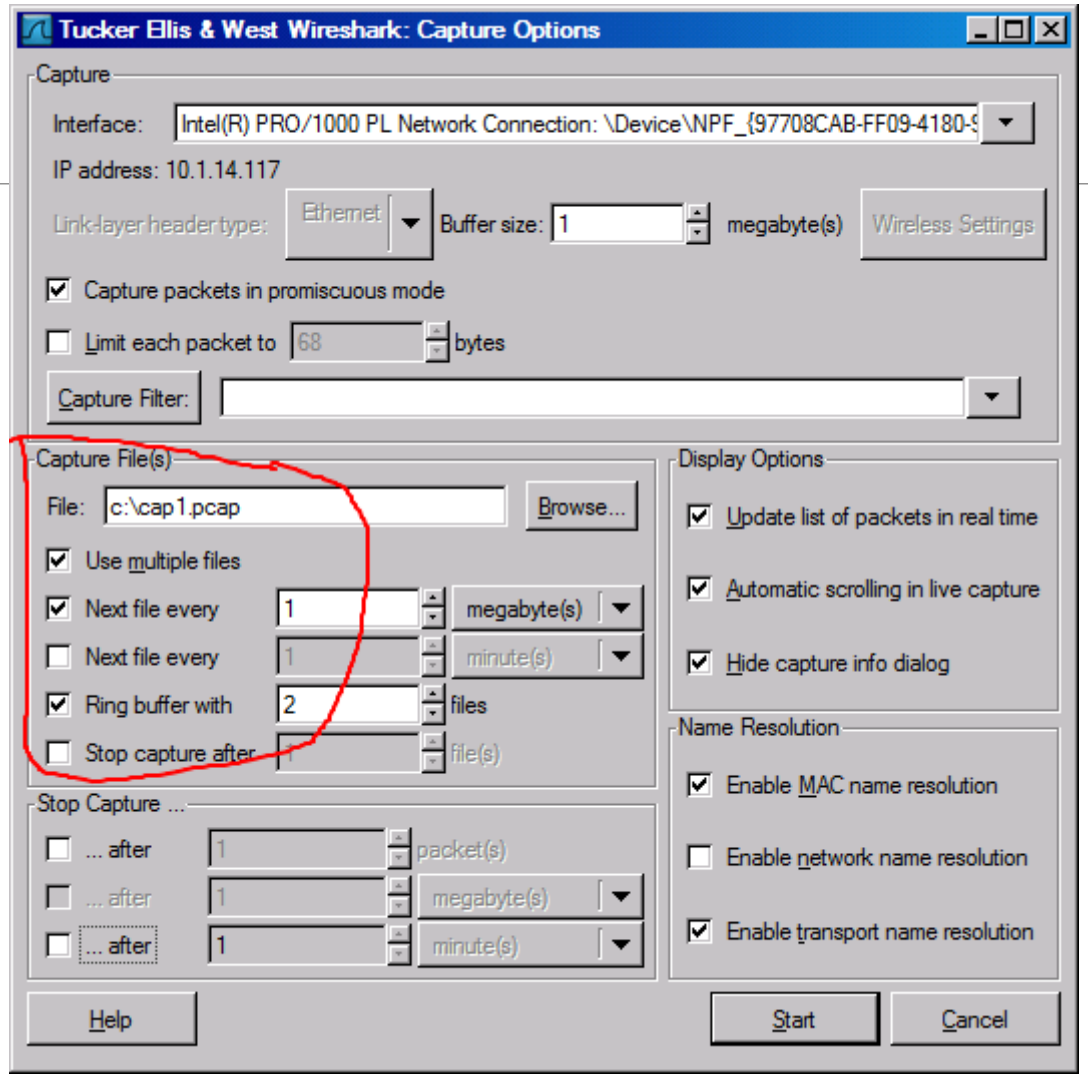
ip

not broadcast not multicast

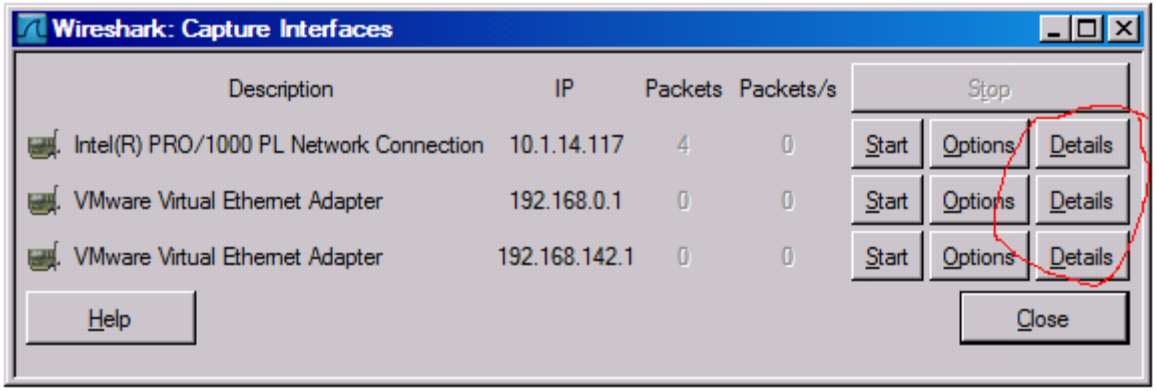
ether host 00:04:13:00:09:a3

Capture Buffer Usage

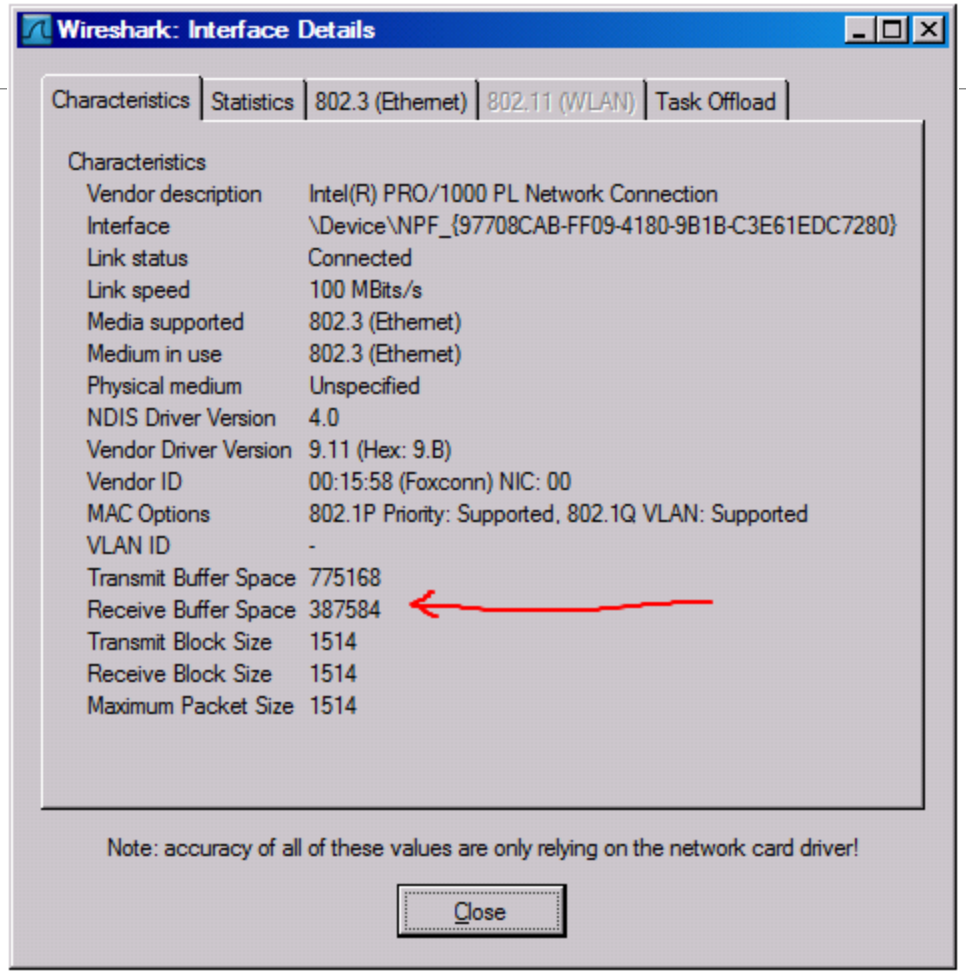



Tucker Ellis & West Wireshark: Capture Options
Capture
Interface:
Intel(R) PRO/1000 PL Network Connection: \Device\NPF_{97708CAB-FF09-4180-9
IP address: 10.1.14.117
Link-layer header type:
Ethernet
Buffer size:
1
megabyte(s)
Wireless Settings
Capture packets in promiscuous mode
Limit each packet to
68
bytes
Capture Filter:
Capture File(s)
File:
c:\cap1.pcap
Browse...
Use multiple files
Next file every
1
megabyte(s)
Next file every
1
minute(s)
Ring buffer with
2
files
Stop capture after
1
file(s)
Display Options
Update list of packets in real time
Automatic scrolling in live capture
Hide capture info dialog
Name Resolution
Enable MAC name resolution
Enable network name resolution
Enable transport name resolution
Stop Capture ...
... after
1
packet(s)
... after
1
megabyte(s)
... after
1
minute(s)
Help
Start
Cancel

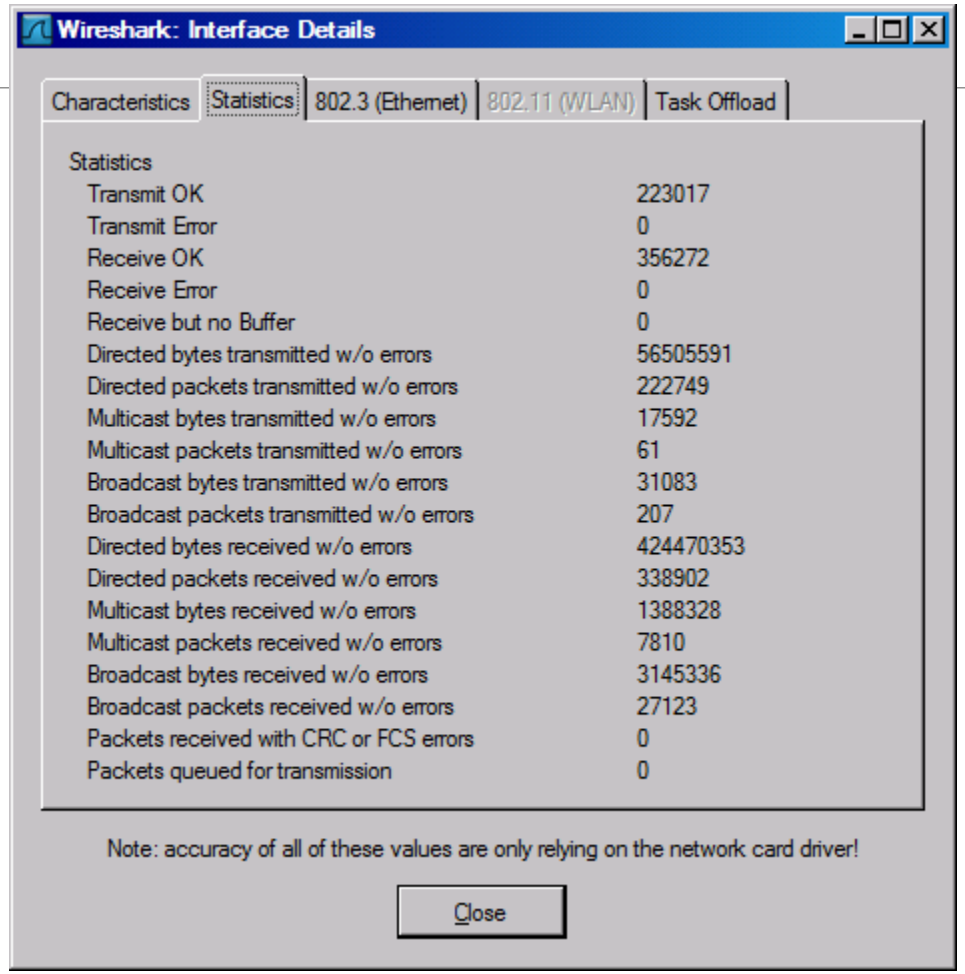
Capture Interfaces



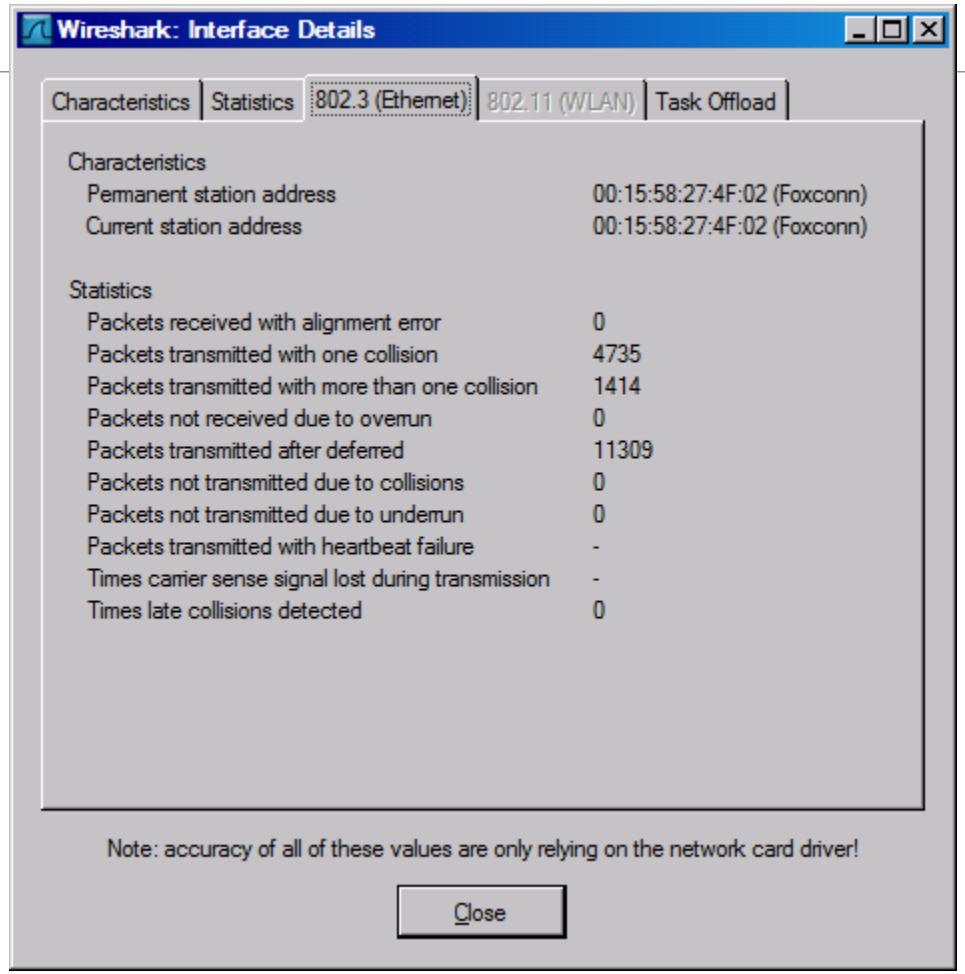
Interface Details: Characteristics



Interface Details: Statistics



Interface Details: 802.3 (Ethernet)

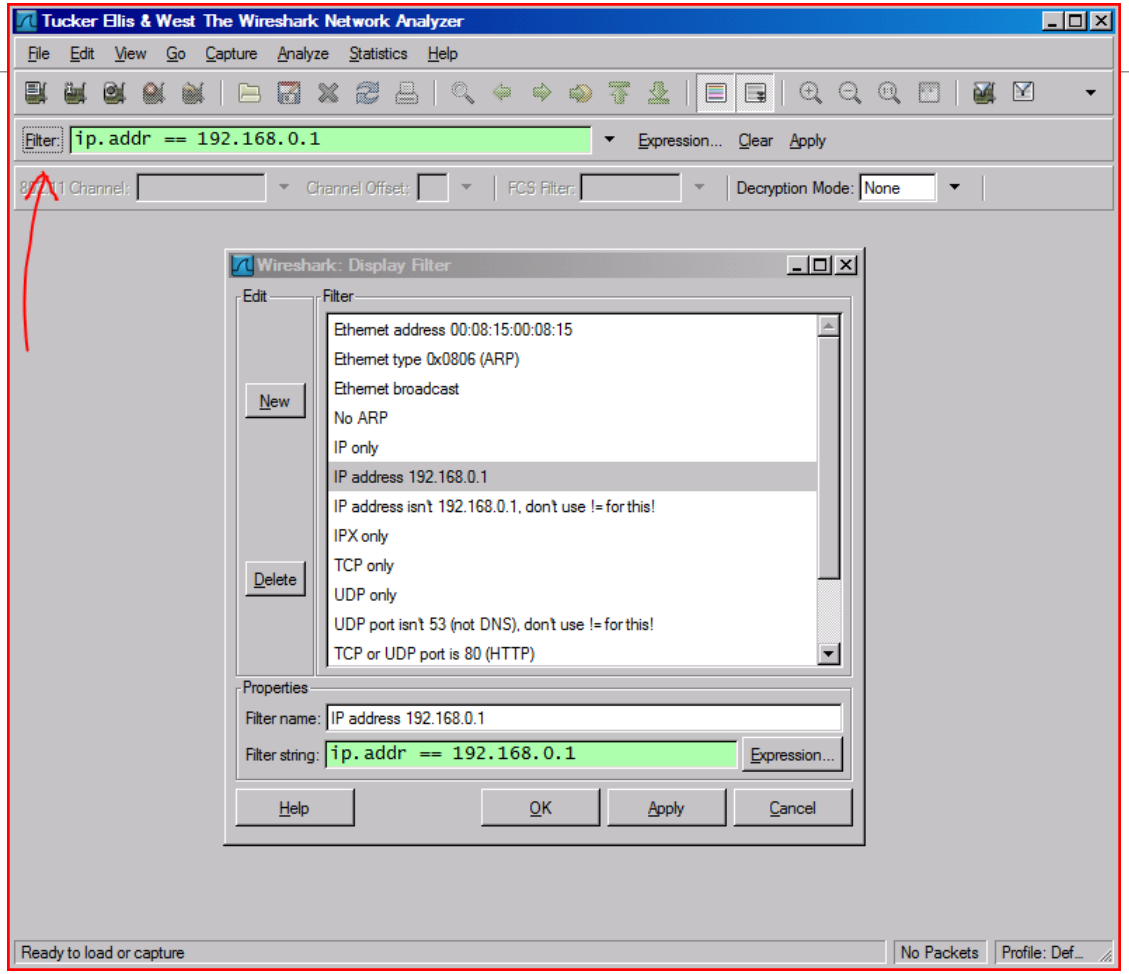


Display Filters (Post-Filters)

Display filters (also called post-filters) only filter the view of what you are seeing. All packets in the capture still exist in the trace

Display filters use their own format and are much more powerful then capture filters

Display Filter



Display Filter Examples

ip.src==10.1.11.00/24

ip.addr==192.168.1.10 && ip.addr==192.168.1.20

tcp.port==80 || tcp.port==3389

!(ip.addr==192.168.1.10 && ip.addr==192.168.1.20)

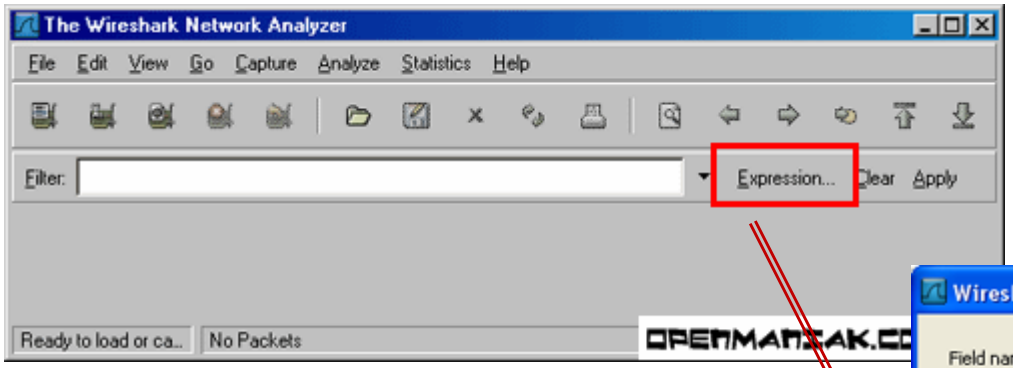
(ip.addr==192.168.1.10 && ip.addr==192.168.1.20) && (tcp.port==445 || tcp.port==139)

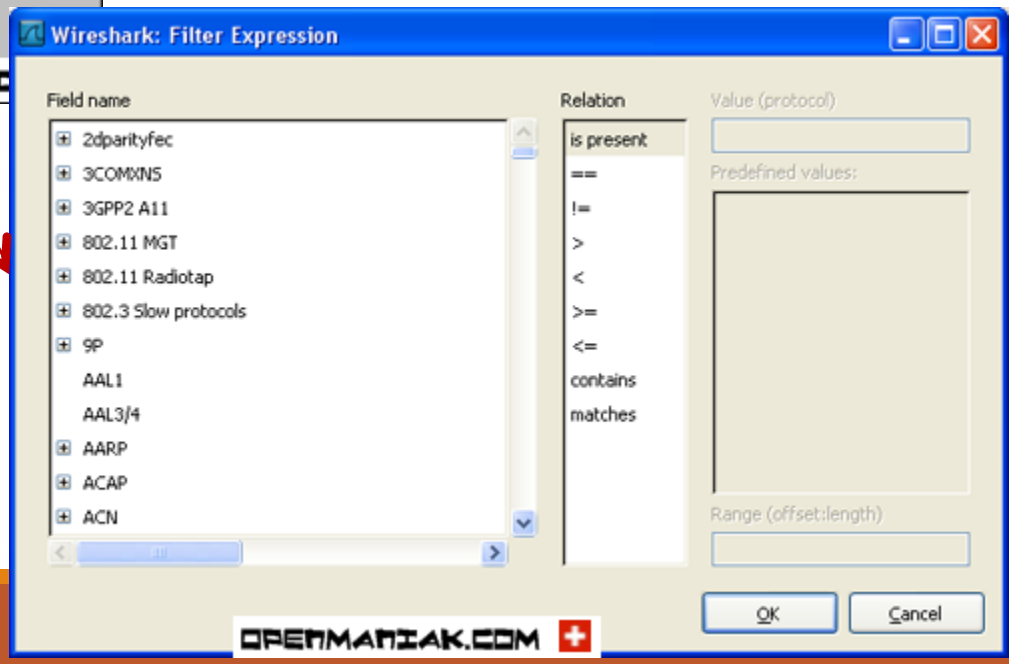
(ip.addr==192.168.1.10 && ip.addr==192.168.1.20) && (udp.port==67 || udp.port==68)

tcp.dstport == 80

Display Filter

Syntax:	**Protocol** .	**String 1** .	**String 2**	**Comparison operator**	**Value**	**Logical Operations**	**Other expression**
Example:	ftp	passive	ip	==	10.2.3.4	xor	icmp.type





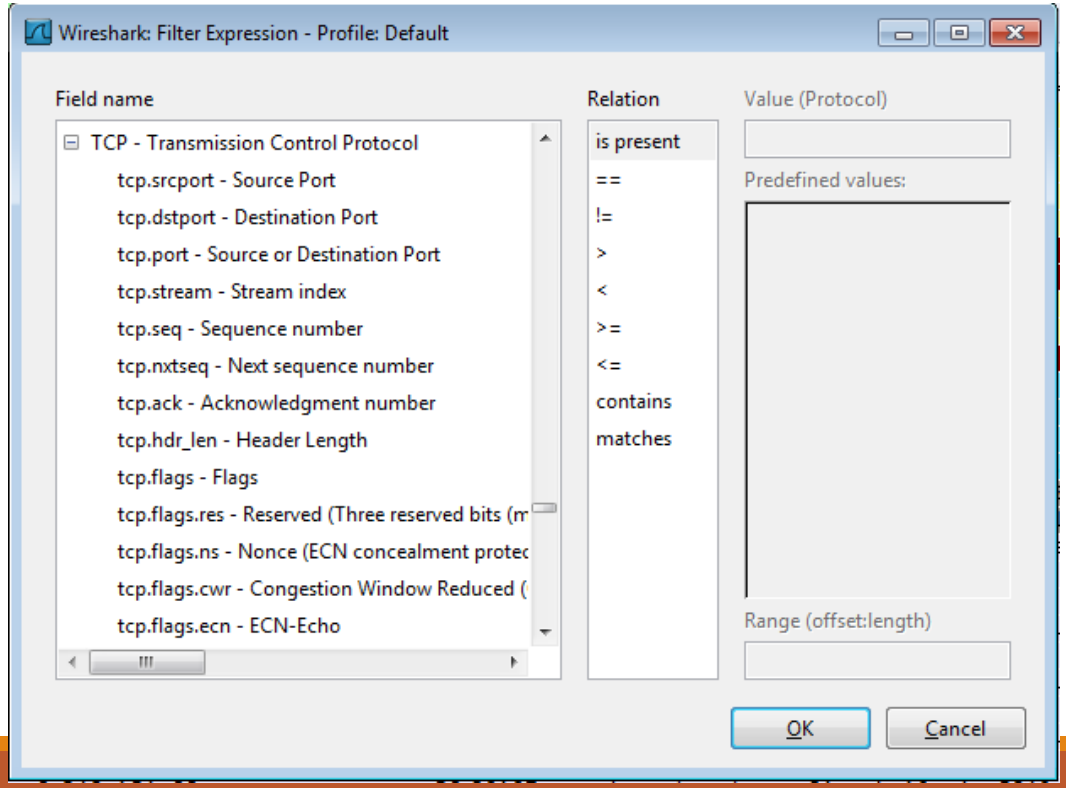
Display Filter

String1, String2 (Optional settings):

- Sub protocol categories inside the protocol.
- Look for a protocol and then click on the "+" character.
- Example:
- **tcp.srcport == 80**
- **tcp.flags == 2**
  - **SYN packet**
  - **Tcp.flags.syn==1**
- **tcp.flags == 18**
  - **SYN/ACK**

- **Note of TCP Flag field:**

U	A	P	R	S	F
R	C	S	S	Y	I
G	K	H	T	N	N



Display Filter Expressions

snmp || dns || icmp

- Display the SNMP or DNS or ICMP traffics.

tcp.port == 25

- Display packets with TCP source or destination port 25.

tcp.flags

- Display packets having a TCP flags

tcp.flags.syn == 0x02

- Display packets with a TCP SYN flag.

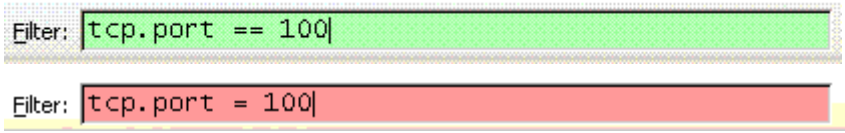
Six comparison operators are available:

English format:	C like format:	Meaning:
eq	==	Equal
ne	!=	Not equal
gt	>	Greater than
lt	<	Less than
ge	>=	Greater or equal
le	<=	Less or equal

Logical expressions:

English format:	C like format:	Meaning:
and	&&	Logical AND
or	||	Logical OR
xor	^^	Logical XOR
not	!	Logical NOT

If the filter syntax is correct, it will be highlighted in green, otherwise if there is a syntax mistake it will be highlighted in red.



Correct syntax

Wrong syntax

Save Filtered Packets After Using Display Filter

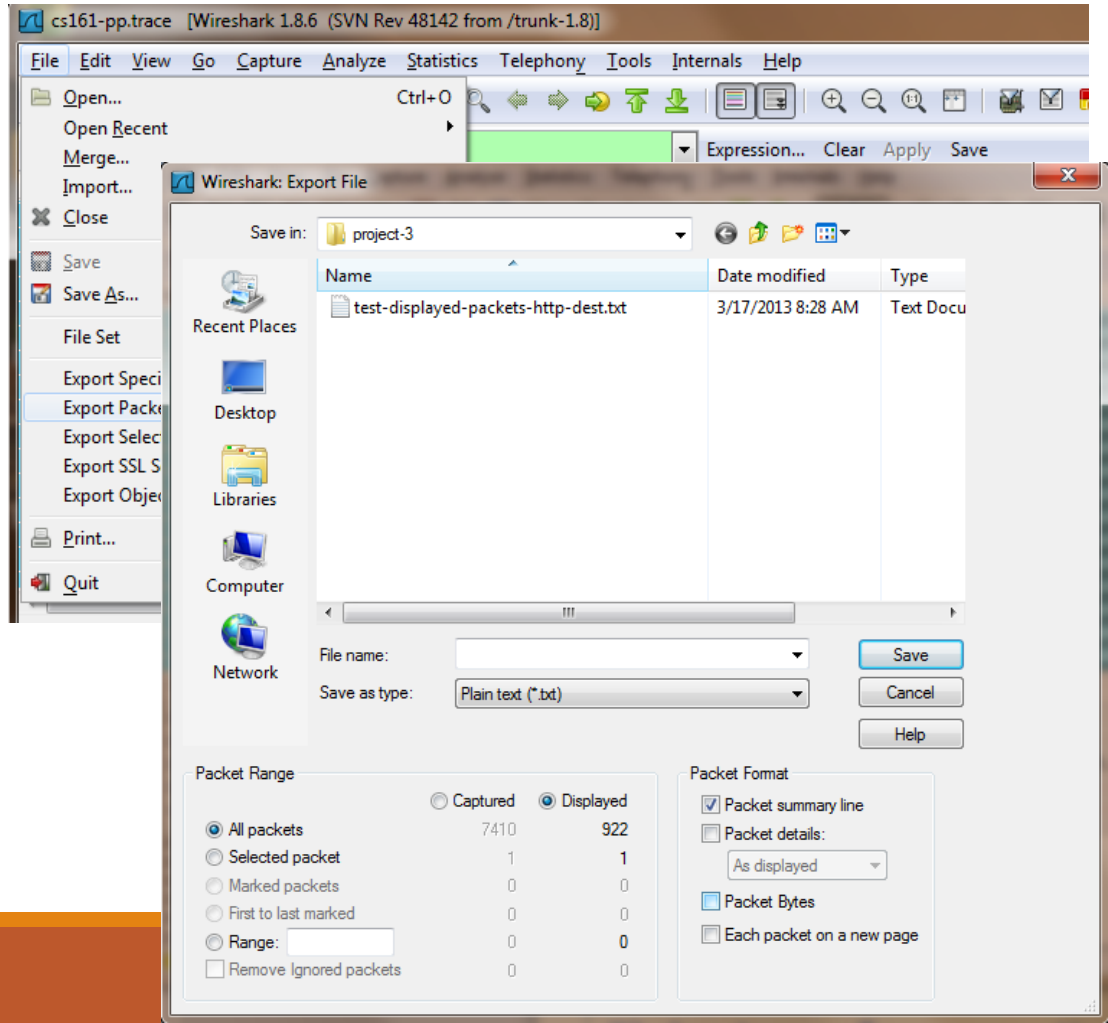
We can also save all filtered packets in text file for further analysis

Operation:

File→Export packet dissections →as "plain text" file

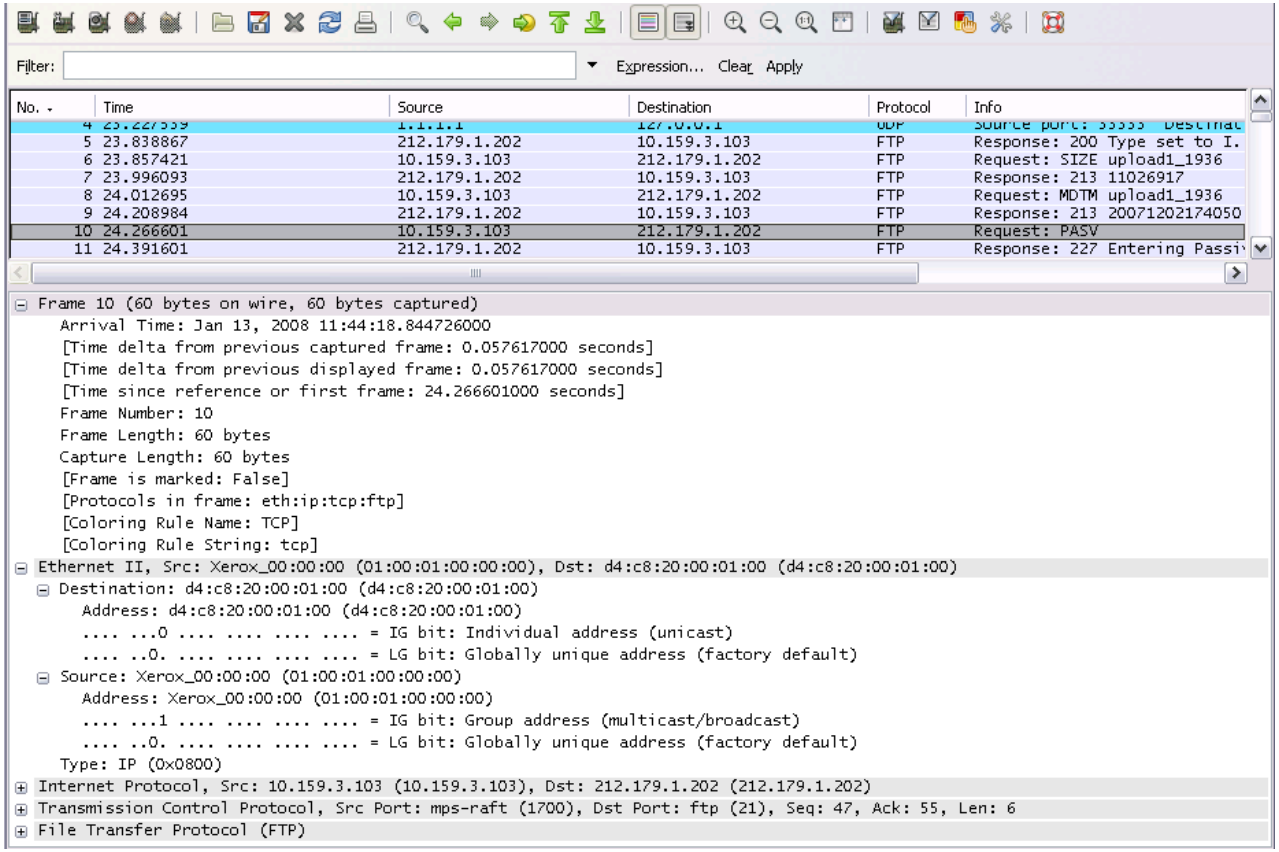
1). In "packet range" option, select "Displayed"

2). In choose "summary line" or "detail"



Analyzing Packets (1/9)

- **Ethernet Frame Example**



Analyzing Packets (2/9)

No.	Time	Source	Destination	Protocol	Info
4	23.227539	1.1.1.1	127.0.0.1	UDP	Source port: 33333 Destinat
5	23.838867	212.179.1.202	10.159.3.103	FTP	Response: 200 Type set to I.
6	23.857421	10.159.3.103	212.179.1.202	FTP	Request: SIZE upload1_1936
7	23.996093	212.179.1.202	10.159.3.103	FTP	Response: 213 11026917
8	24.012695	10.159.3.103	212.179.1.202	FTP	Request: MDTM upload1_1936
9	24.208984	212.179.1.202	10.159.3.103	FTP	Response: 213 20071202174050
10	24.266601	10.159.3.103	212.179.1.202	FTP	Request: PASV

```
⊞ Frame 10 (60 bytes on wire, 60 bytes captured)
⊞ Ethernet II, Src: Xerox_00:00:00 (01:00:01:00:00:00), Dst: d4:c8:20:00:01:00 (d4:c8:20:00:01:00)
⊟ Internet Protocol, Src: 10.159.3.103 (10.159.3.103), Dst: 212.179.1.202 (212.179.1.202)
    Version: 4
    Header length: 20 bytes
  ⊟ Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00)
      0000 00.. = Differentiated Services Codepoint: Default (0x00)
      .... ..0. = ECN-Capable Transport (ECT): 0
      .... ...0 = ECN-CE: 0
    Total Length: 46
    Identification: 0x5f49 (24393)
  ⊟ Flags: 0x04 (Don't Fragment)
      0... = Reserved bit: Not set
      .1.. = Don't fragment: Set
      ..0. = More fragments: Not set
    Fragment offset: 0
    Time to live: 128
    Protocol: TCP (0x06)
  ⊟ Header checksum: 0xb6fd [correct]
      [Good: True]
      [Bad : False]
    Source: 10.159.3.103 (10.159.3.103)
    Destination: 212.179.1.202 (212.179.1.202)
⊞ Transmission Control Protocol, Src Port: mps-raft (1700), Dst Port: ftp (21), Seq: 47, Ack: 55, Len: 6
⊞ File Transfer Protocol (FTP)
```

Analyzing Packets (3/9)

Filter: Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
9	24.208984	212.179.1.202	10.159.3.103	FTP	Response: 213 20071202174050
10	24.266601	10.159.3.103	212.179.1.202	FTP	Request: PASV
11	24.391601	212.179.1.202	10.159.3.103	FTP	Response: 227 Entering Passi

```
⊞ Frame 10 (60 bytes on wire, 60 bytes captured)
⊞ Ethernet II, Src: Xerox_00:00:00 (01:00:01:00:00:00), Dst: d4:c8:20:00:01:00 (d4:c8:20:00:01:00)
⊞ Internet Protocol, Src: 10.159.3.103 (10.159.3.103), Dst: 212.179.1.202 (212.179.1.202)
⊟ Transmission Control Protocol, Src Port: mps-raft (1700), Dst Port: ftp (21), Seq: 47, Ack: 55, Len: 6
    Source port: mps-raft (1700)
    Destination port: ftp (21)
    [Stream index: 1]
    Sequence number: 47    (relative sequence number)
    [Next sequence number: 53    (relative sequence number)]
    Acknowledgement number: 55    (relative ack number)
    Header length: 20 bytes
  ⊟ Flags: 0x18 (PSH, ACK)
      0... .... = Congestion Window Reduced (CWR): Not set
      .0.. .... = ECN-Echo: Not set
      ..0. .... = Urgent: Not set
      ...1 .... = Acknowledgement: Set
      .... 1... = Push: Set
      .... .0.. = Reset: Not set
      .... ..0. = Syn: Not set
      .... ...0 = Fin: Not set
    Window size: 16945
  ⊟ Checksum: 0x8b8d [validation disabled]
      [Good Checksum: False]
      [Bad Checksum: False]
  ⊟ [SEQ/ACK analysis]
      [This is an ACK to the segment in frame: 9]
      [The RTT to ACK the segment was: 0.057617000 seconds]
      [Number of bytes in flight: 6]
⊞ File Transfer Protocol (FTP)
```

Analyzing Packets (4/9)

(Untitled) - Wireshark

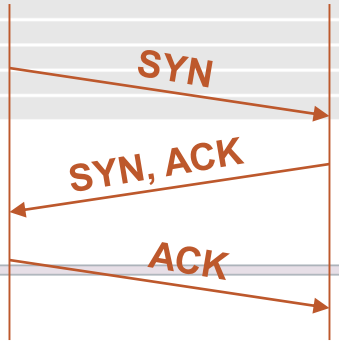
File Edit View Go Capture Analyze Statistics Telephony Tools Help

Filter: Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
1	0.000000	192.168.2.100	10.40.41.2	ICMP	Echo (ping) request
2	2.183304	192.168.2.100	10.40.41.2	ICMP	Echo (ping) request
3	3.430100	192.168.2.100	212.150.49.10	DNS	Standard query A www.ynet.co.il
4	3.457181	212.150.49.10	192.168.2.100	DNS	Standard query response CNAME ynet.co.il.d4p.net CNAME a39.g.
5	3.461602	192.168.2.100	212.150.49.10	DNS	Standard query A www.lenovo.com
6	3.623867	192.168.2.100	212.143.162.157	TCP	dzdaemon > http [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=1 TSV
7	3.728385	212.143.162.157	192.168.2.100	TCP	http > dzdaemon [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=145
8	3.728429	192.168.2.100	212.143.162.157	TCP	dzdaemon > http [ACK] Seq=1 Ack=1 Win=128480 Len=0
9	3.728839	192.168.2.100	212.143.162.157	HTTP	GET / HTTP/1.1
10	3.768896	212.143.162.157	192.168.2.100	TCP	http > dzdaemon [ACK] Seq=1 Ack=580 Win=6948 Len=0
11	3.770703	212.143.162.157	192.168.2.100	HTTP	HTTP/1.0 301 Moved Permanently
12	3.772411	192.168.2.100	212.143.162.157	HTTP	GET /home/0,7340,L-8,00.html HTTP/1.1

```
⊞ Frame 5 (74 bytes on wire, 74 bytes captured)
⊞ Ethernet II, Src: IntelCor_a2:d8:9a (00:1c:bf:a2:d8:9a), Dst: EdimaxTe_6e:2f:7d (00:0e:2e:6e:2f:7d)
⊞ Internet Protocol, Src: 192.168.2.100 (192.168.2.100), Dst: 212.150.49.10 (212.150.49.10)
⊞ User Datagram Protocol, Src Port: natuslink (2895), Dst Port: domain (53)
⊞ Domain Name System (query)
```

```
0000  00 0e 2e 6e 2f 7d 00 1c  bf a2 d8 9a 08 00 45 00   ...n/}.. ......E.
0010  00 3c 7f ea 00 00 80 11  f2 19 c0 a8 02 64 d4 96   .<...... .....d..
0020  31 0a 0b 4f 00 35 00 28  f5 df 9e d7 01 00 00 01   1..O.5.( ........
0030  00 00 00 00 00 00 03 77  77 77 06 6c 65 6e 6f 76   .......w ww.lenov
0040  6f 03 63 6f 6d 00 00 01  00 01                      o.com... ..
```



File: "C:\DOCUME~1\yoram\LOCALS~1\Temp\wi... | Packets: 1303 Displayed: 1303 Marked: 0 Dropped: 0 | Profile: Default

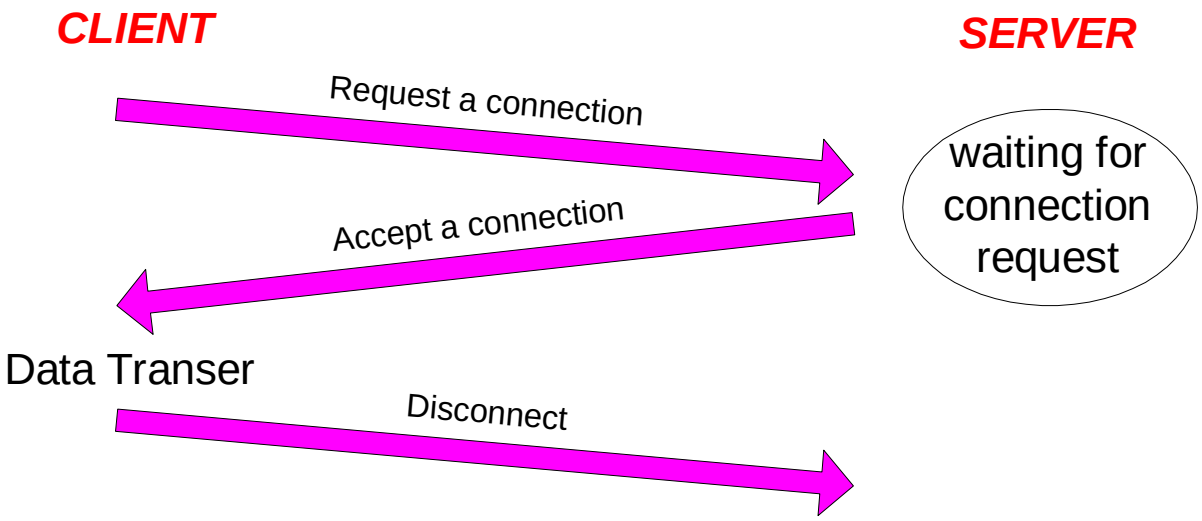
Connection-Oriented

Before any data transfer, TCP establishes a **connection**:

- One TCP entity is waiting for a connection ("**server**")
- The other TCP entity ("**client**") contacts the server

The actual procedure for setting up connections is more complex.

Each connection is full duplex



Reliable

Byte stream is broken up into chunks which are called **segments**

Receiver sends acknowledgements (ACKs) for segments

TCP maintains a timer. If an ACK is not received in time, the segment is retransmitted

Detecting errors:

TCP has checksums for header and data. Segments with invalid checksums are discarded

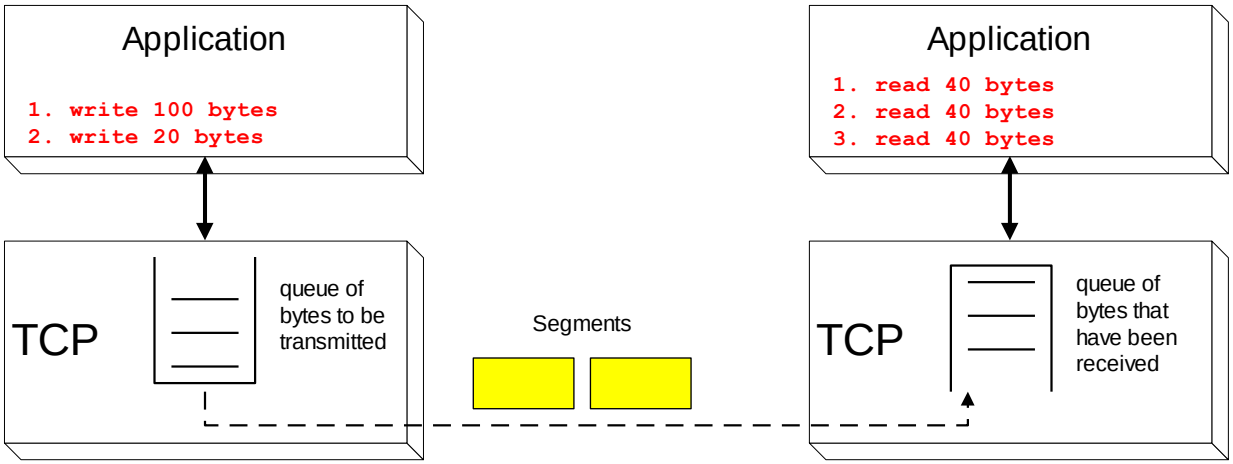
Each byte that is transmitted has a sequence number

Byte Stream Service

To the lower layers, TCP handles data in blocks, the segments.

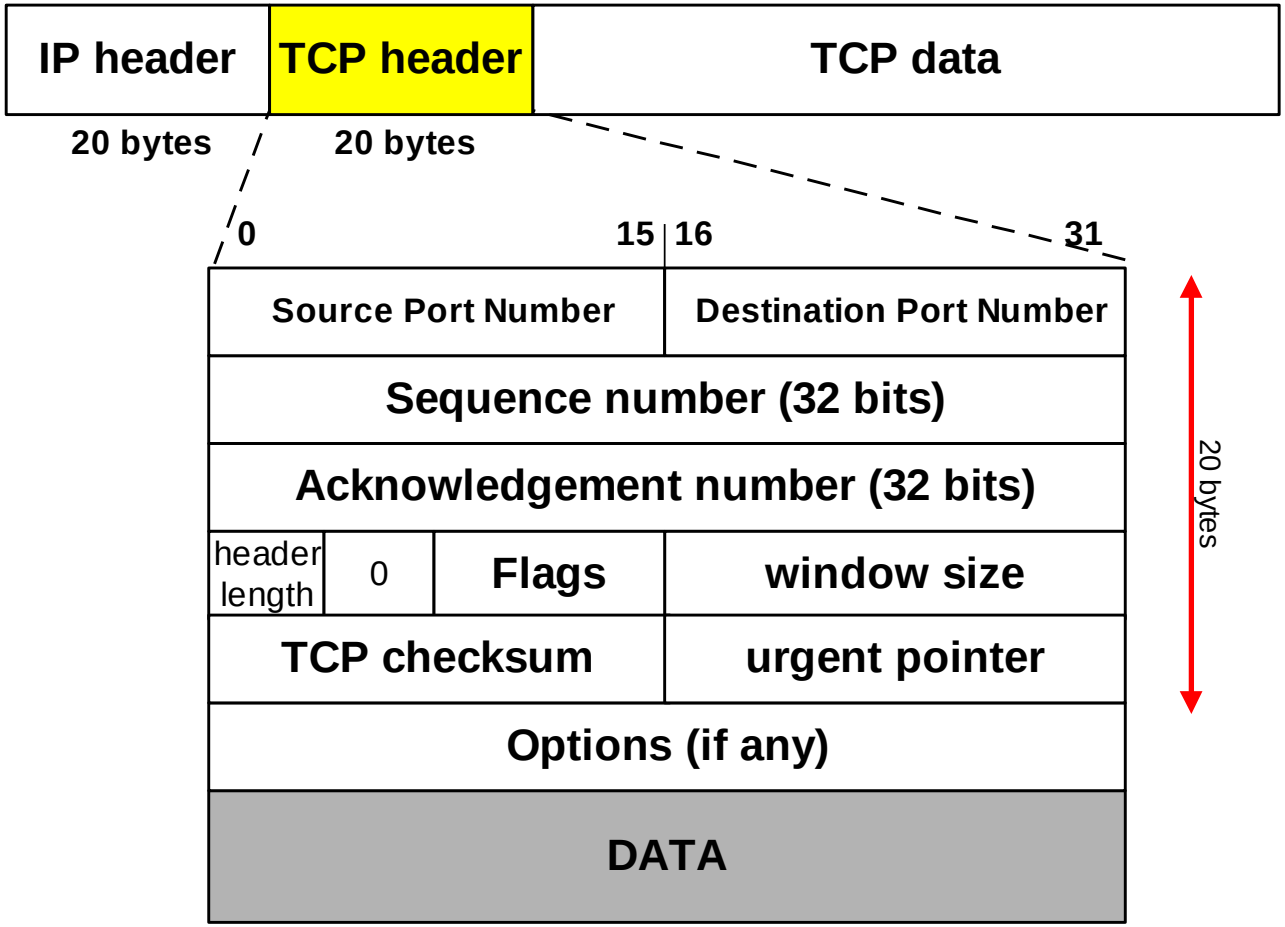
To the higher layers TCP handles data as a sequence of bytes and does not identify boundaries between bytes

So: Higher layers do not know about the beginning and end of segments !



TCP Format

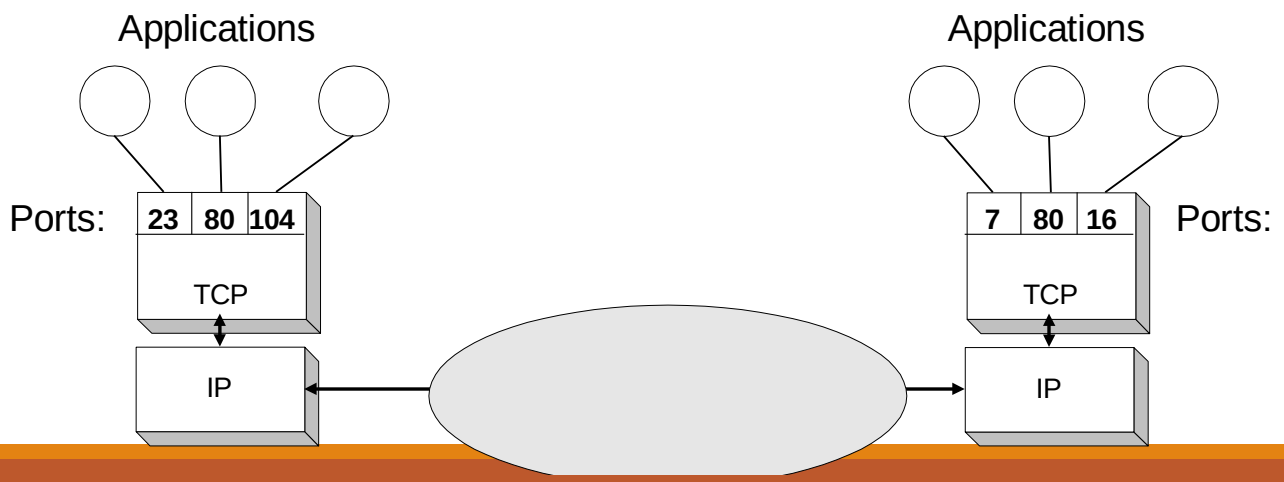
TCP segments have a 20 byte header with >= 0 bytes of data.



TCP header fields

Port Number:

- A port number identifies the endpoint of a connection.
- A pair `<IP address, port number>` identifies one endpoint of a connection.
- Two pairs `<client IP address, server port number>` and `<server IP address, server port number>` identify a TCP connection.



TCP header fields

Sequence Number (SeqNo):

- Sequence number is 32 bits long.
- So the range of SeqNo is

  $0 <= SeqNo <= 2^{32} - 1 \approx 4.3$ Gbyte
- Each sequence number identifies a byte in the byte stream
- Initial Sequence Number (ISN) of a connection is set during connection establishment

TCP header fields

Acknowledgement Number (AckNo):

- Acknowledgements are piggybacked, I.e

  a segment from A -> B can contain an acknowledgement for a data sent in the B -> A direction

- A hosts uses the AckNo field to send acknowledgements. (If a host sends an AckNo in a segment it sets the "**ACK flag**")
- The AckNo contains the next SeqNo that a hosts wants to receive
  Example: The acknowledgement for a segment with sequence numbers 0-1500 is AckNo=1501

TCP header fields

Acknowledge Number (cont'd)

- TCP uses the sliding window flow protocol (see CS 457) to regulate the flow of traffic from sender to receiver
- TCP uses the following variation of sliding window:
  - no NACKs (**N**egative **ACK**nowledgement)
  - only cumulative ACKs

Example:

Assume: Sender sends two segments with "1..1500" and "1501..3000", but receiver only gets the second segment.

In this case, the receiver cannot acknowledge the second packet. It can only send AckNo=1

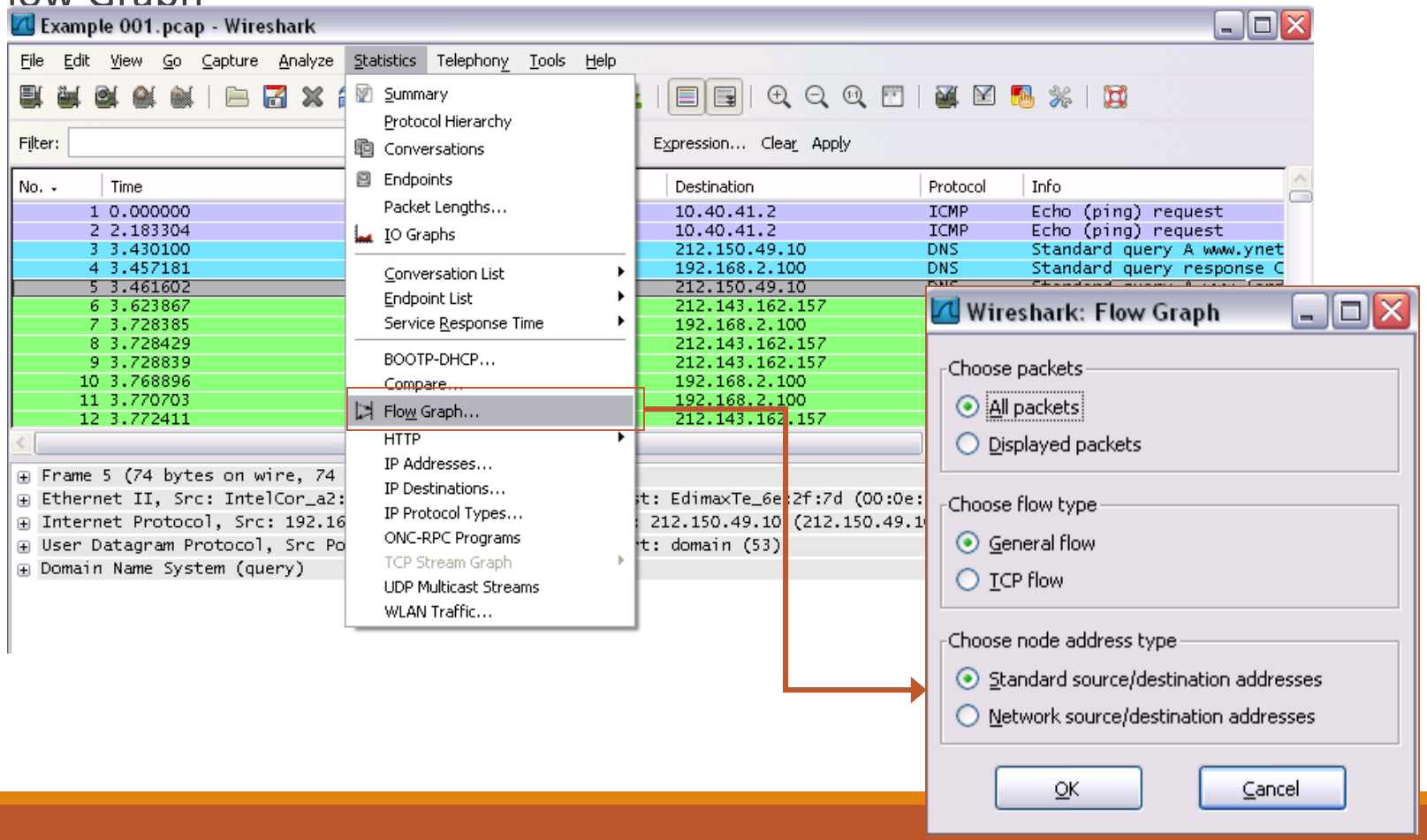
TCP header fields

Header Length (4bits):

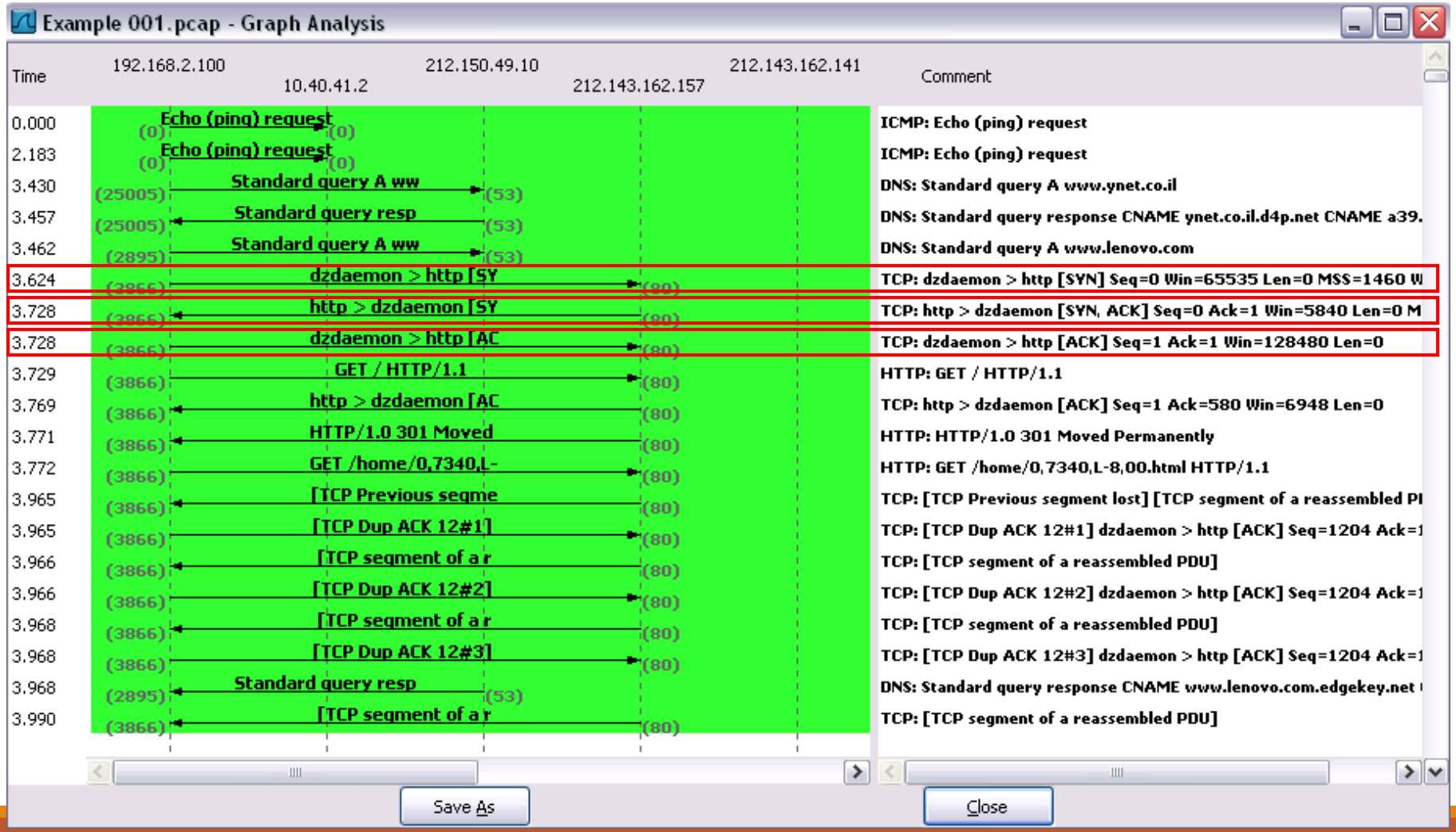
- Length of header in 32-bit words
- Note that TCP header has variable length (with minimum 20 bytes)

Analyzing Packets (5/9)

Flow Graph



Analyzing Packets (6/9)



Analyzing Packets (7/9)

Snif2 --- HTTP Example.cap - Wireshark

File Edit View Go Capture Analyze Statistics Telephony Tools Help

Filter: Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
38	10.031657	10.115.243.30	10.114.30.180	DNS	Standard query response CNAME toolbarqueries.l.google.
39	10.032397	10.114.30.180	64.233.183.99	TCP	peport > http [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=
40	10.035182	64.233.183.99	10.114.30.180	TCP	http > peport [SYN, ACK] Seq=0 Ack=1 Win=32768 Len=0 M
41	10.035234	10.114.30.180	64.233.183.99	TCP	peport > http [ACK] Seq=1 Ack=1 Win=131400 Len=0 TSV=1
42	10.035443	10.114.30.180	64.233.183.99	HTTP	nt-auto&ch=6174981939&freshn
43	10.145399	64.233.183.99	10.114.30.180	TCP	Ack=447 Win=32768 Len=0 TSV=
44	10.312738	64.233.183.99	10.114.30.180	TCP	led PDU]
45	10.312814	64.233.183.99	10.114.30.180	HTTP	])
46	10.312862	10.114.30.180	64.233.183.99	TCP	7 Ack=322 Win=131076 Len=0 T
47	10.399875	10.114.30.180	10.115.243.30	DNS	tcp.dc._msdcs.ndi.local
48	10.400571	10.115.243.30	10.114.30.180	DNS	o such name

- Frame 42 (512 bytes on wire, 512 bytes captured)
- Ethernet II, Src: Ibm_42:c2:4d (00:09:6b:42:c2:4d), Dst: LucentTe_cf:cd:2c (00:30:6d:cf:cd:2c)
- Internet Protocol, Src: 10.114.30.180 (10.114.30.180), Dst: 64.233.183.99 (64.233.183.99)
- Transmission Control Protocol, Src Port: peport (1449), Dst Port: http (80), Seq: 1, Ack: 1, L
- Hypertext Transfer Protocol

```
0000  00 30 6d cf cd 2c 00 09  6b 42 c2 4d 08 00 45 00   .0m..,.. kB.M..E.
0010  01 f2 1a 79 40 00 80 06  bd 1a 0a 72 1e b4 40 e9   ...y@... ...r..@.
0020  b7 63 05 a9 00 50 f2 b3  c2 a3 34 52 27 7f 80 18   .c...P.. ..4R'...
0030  80 52 3a 66 00 00 01 01  08 0a 00 01 c7 f3 90 77   .R:f.... .......w
0040  be 82 47 45 54 20 2f 73  65 61 72 63 68 3f 63 6c   ..GET /s earch?cl
0050  69 65 6e 74 3d 6e 61 76  63 6c 69 65 6e 74 2d 61   ient=nav client-a
```

File: "D:\Customers\Examples\Snif2 --- HTTP Exa... | Packets: 1648 Displayed: 1648 Marked: 0

Mark Packet (toggle)
Set Time Reference (toggle)
Apply as Filter
Prepare a Filter
Conversation Filter
Colorize Conversation
SCTP
Follow TCP Stream
Follow UDP Stream
Follow SSL Stream

Follow TCP Stream

Stream Content

```
GET /search?client=navclient-auto&ch=6174981939&freshness_check=4ilp-
GrPqKEx_r_lNxaYw&iqrn=qr4&orig=0J&ie=UTF-8&oe=UTF-8&features=Rank&q=info:http%3A%2F%2Fwww%2Eynet%2Eco%2Eil%2F
HTTP/1.1
User-Agent: Mozilla/4.0 (compatible; GoogleToolbar 2.0.114.9-big; Windows XP 5.1)
Host: toolbarqueries.google.com
Cache-Control: no-cache
Cookie: PREF=ID=1a18560743a17669:TB=2:CR=1:TM=1113765996:LM=1119978279:GM=1:S=7NmjkcGkIc845ngM; rememberme=false

HTTP/1.1 200 OK
Transfer-Encoding: chunked
Date: Mon, 11 Jul 2005 08:21:03 GMT
Content-Type: text/html
Cache-Control: private
Server: GWS/2.1
Via: 1.1 cache1 (NetCache NetApp/5.5R2D5), Version 2.0-Build_Linux_1336 $Date: 04/13/2005 15:53:0038$(IWSS), 1.1
cache1 (NetCache NetApp/5.5R2D5)

e
```

Find | Save As | Print | Entire conversation (767 bytes) | ASCII | EBCDIC | Hex Dump | C Arrays | Raw

Help | Filter Out This Stream | Close

Analyzing Packets (8/9)

Snif2 --- HTTP Example.cap - Wireshark

File Edit View Go Capture Analyze Statistics Telephony Tools Help

Filter: (tcp.stream eq 5) Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
39	10.032397	10.114.30.180	64.233.183.99	TCP	peport > http [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=
40	10.035182	64.233.183.99	10.114.30.180	TCP	http > peport [SYN, ACK] Seq=0 Ack=1 Win=32768 Len=0 M
41	10.035234	10.114.30.180	64.233.183.99	TCP	peport > http [ACK] Seq=1 Ack=1 Win=131400 Len=0 TSV=1
42	10.035443	10.114.30.180	64.233.183.99	HTTP	GET /search?client=navclient-auto&ch=6174981939&freshn
43	10.145399	64.233.183.99	10.114.30.180	TCP	http > peport [ACK] Seq=1 Ack=447 Win=32768 Len=0 TSV=
44	10.312738	64.233.183.99	10.114.30.180	TCP	[TCP segment of a reassembled PDU]
45	10.312814	64.233.183.99	10.114.30.180	HTTP	HTTP/1.1 200 OK (text/html)
46	10.312862	10.114.30.180	64.233.183.99	TCP	peport > http [ACK] Seq=447 Ack=322 Win=131076 Len=0 T
169	20.311539	64.233.183.99	10.114.30.180	TCP	http > peport [FIN, ACK] Seq=322 Ack=447 Win=32768 Len
170	20.311629	10.114.30.180	64.233.183.99	TCP	peport > http [ACK] Seq=447 Ack=323 Win=131076 Len=0 T
192	21.479689	10.114.30.180	64.233.183.99	TCP	peport > http [FIN, ACK] Seq=447 Ack=323 Win=131076 Le
194	21.480926	64.233.183.99	10.114.30.180	TCP	http > peport [ACK] Seq=323 Ack=448 Win=32768 Len=0 TS

```
⊞ Frame 39 (78 bytes on wire, 78 bytes captured)
⊞ Ethernet II, Src: Ibm_42:c2:4d (00:09:6b:42:c2:4d), Dst: LucentTe_cf:cd:2c (00:30:6d:cf:cd:2c)
⊞ Internet Protocol, Src: 10.114.30.180 (10.114.30.180), Dst: 64.233.183.99 (64.233.183.99)
⊞ Transmission Control Protocol, Src Port: peport (1449), Dst Port: http (80), Seq: 0, Len: 0
```

```
0010  00 40 1a 77 40 00 80 06  be ce 0a 72 1e b4 40 e9   .@.w@... ...r..@.
0020  b7 63 05 a9 00 50 f2 b3  c2 a2 00 00 00 00 b0 02   .c...P.. ........
0030  ff ff 59 3d 00 00 02 04  05 b4 01 03 03 02 01 01   ..Y=.... ........
0040  08 0a 00 00 00 00 00 00  00 00 01 01 04 02         ........ ......
```

File: "D:\Customers\Examples\Snif2 --- HTTP Ex... | Packets: 1648 Displayed: 12 Marked: 0 | Profile: Default

Analyzing Packets (9/9)

RTP Example - SIP - 02.pcap - Wireshark

File Edit View Go Capture Analyze Statistics Telephony Tools Help

Telephony: ANSI, Fax T38 Analysis..., GSM, H.225..., IAX2, ISUP Messages..., LTE MAC..., MTP3, RTP, SCTP, SIP..., SMPP Operations..., UCP Messages..., VoIP Calls, WAP-WSP...

RTP: Show All Streams, Stream Analysis...

Wireshark: RTP Stream Analysis

Forward Direction | Reversed Direction

Analysing stream from 192.168.2.100 port 5004 to 78.136.29.109 port 6062 SSRC = 0x39FF6709

Packet	Sequence	Delta(ms)	Filtered Jitter(ms)	Skew(ms)	IP BW(kbps)	Marker	Status
1417	19063	0.00	0.00	0.00	24.48		[Ok]
1419	19064	0.00	0.00	0.00	24.48		[Ok]
1421	19065	0.00	0.00	0.00	24.48		[Ok]
1423	19066	0.00	0.00	0.00	24.48		[Ok]
1425	19067	0.00	0.00	0.00	24.48		[Ok]
1427	19068	0.00	0.00	0.00	24.48		[Ok]
1429	19069	0.00	0.00	0.00	24.48		[Ok]
1431	19070	0.00	0.00	0.00	24.48		[Ok]

Max delta = 0.00 ms at packet no. 0
Max jitter = 0.00 ms. Mean jitter = 0.00 ms.
Max skew = 0.00 ms.
Total RTP packets = 2098 (expected 2098) Lost RTP packets = 0 (0.00%) Sequence errors = 0
Duration 62.85 s (0 ms clock drift, corresponding to 1 Hz (+0.00%)

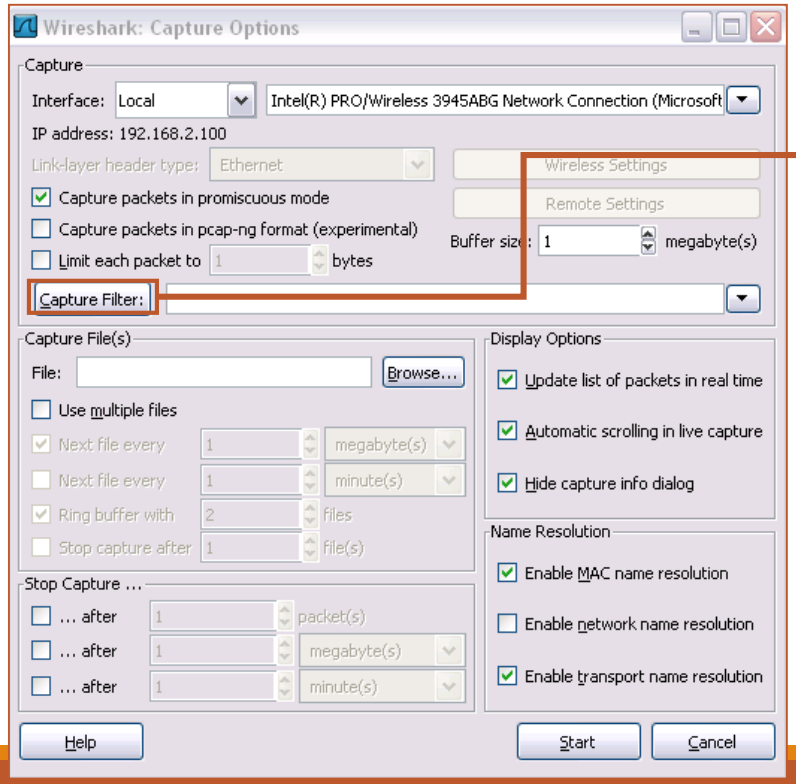
Save payload... | Save as CSV... | Refresh | Jump to | Graph | Next non-Ok | Close

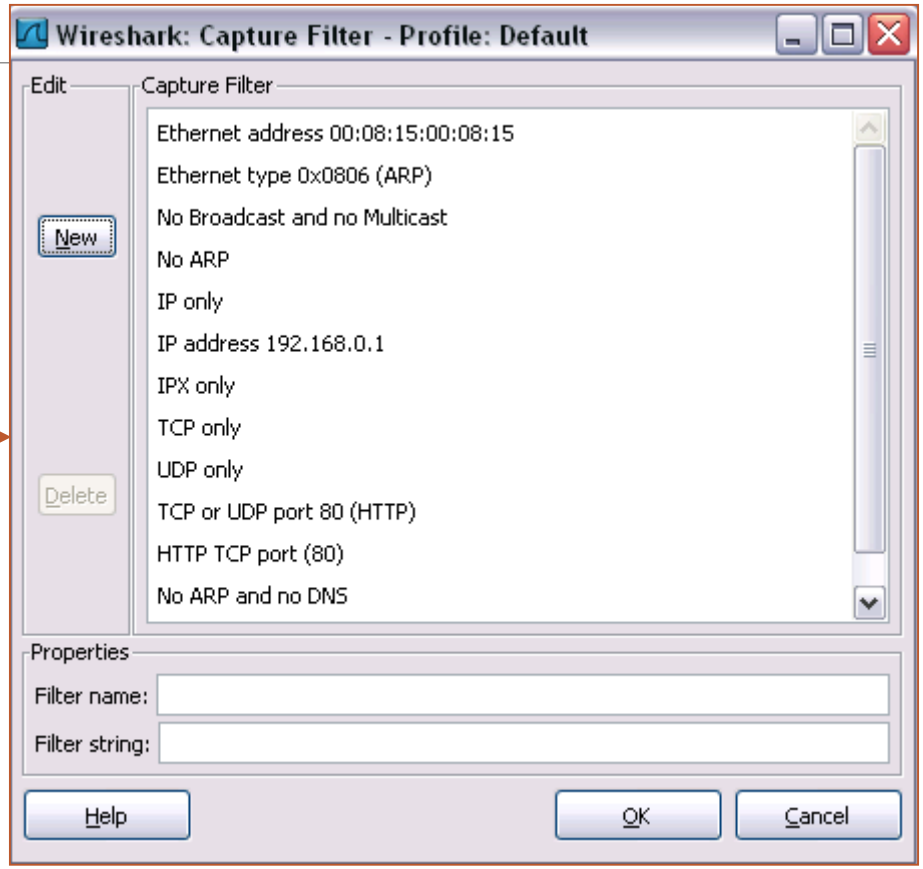
Stable stream BW

Filtering Packets (1/4)

Applying Filter when Capturing Packets

Capture → Interfaces → Options:





Filtering Packets (2/4)

Example 003.pcap - Wireshark

File Edit View Go Capture Analyze Statistics Telephony Tools Help

Filter: Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
485	47.639995	192.168.2.100	[illegible]	TCP	17231 > http [FIN, ACK] Seq=1409 Ack=380 Win=12810
486	47.649881	192.168.2.100	212.143.162.152	TCP	17241 > http [SYN] Seq=0 Win=65535 Len=0 MSS=1460
487	47.666485	212.143.162.152	192.168.2.100	TCP	http > 17237 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0
488	47.666530	192.168.2.100	[illegible]	[illegible]	[illegible]
489	47.666898	192.168.2.100	[illegible]	[illegible]	[illegible]
490	47.667431	192.168.2.100	[illegible]	[illegible]	[illegible]
491	47.675090	212.143.162.1	[illegible]	[illegible]	[illegible]
492	47.675136	192.168.2.100	[illegible]	[illegible]	[illegible]
493	47.677582	212.143.162.1	[illegible]	[illegible]	[illegible]
494	47.677624	192.168.2.100	[illegible]	[illegible]	[illegible]
495	47.677858	192.168.2.100	[illegible]	[illegible]	[illegible]
496	47.695323	212.143.162.1	[illegible]	[illegible]	[illegible]
497	47.697056	212.143.162.1	[illegible]	[illegible]	[illegible]
498	47.737850	212.143.162.1	[illegible]	[illegible]	[illegible]
499	47.739896	212.143.162.1	[illegible]	[illegible]	[illegible]
500	47.788636	212.143.162.1	[illegible]	[illegible]	[illegible]
501	47.797761	192.168.2.100	[illegible]	[illegible]	[illegible]
502	47.826481	212.143.162.1	[illegible]	[illegible]	[illegible]
503	47.826527	192.168.2.100	[illegible]	[illegible]	[illegible]
504	47.826913	192.168.2.100	[illegible]	[illegible]	[illegible]
505	47.839189	212.143.162.1	[illegible]	[illegible]	[illegible]
506	47.841074	212.143.162.1	[illegible]	[illegible]	[illegible]
507	47.856460	212.143.162.1	[illegible]	[illegible]	[illegible]
508	47.858425	212.143.162.1	[illegible]	[illegible]	[illegible]

```
⊞ Frame 485 (54 bytes on wire, 54 bytes captured)
⊞ Ethernet II, Src: IntelCor_a2:d8:9a (00:1c:bf:a2:d
⊞ Internet Protocol, Src: 192.168.2.100 (192.168.2.1
⊞ Transmission Control Protocol, Src Port: 17231 (17

0000  00 0e 2e 6e 2f 7d 00 1c  bf a2 d8 9a 08 00 45 0
0010  00 28 94 52 40 00 80 06  2c 49 c0 a8 02 64 d4 8
0020  a2 98 43 4f 00 50 b6 d6  a4 44 e9 7c 0f 5f 50 1
0030  fa 32 e3 d5 00 00
```

Wireshark: Filter Expression - Profile: Default

Field name	Relation	Value (character string)
⊞ Mesh - Mesh Header	is present	
message/http - Media Type: message/http	==	Predefined values:
⊞ Messenger - Microsoft Messenger Service	!=	
⊞ MGCP - Media Gateway Control Protocol	>	
⊞ MGMT - DCE/RPC Remote Management	<	
⊞ MIKEY - Multimedia Internet KEYing	>=	
⊞ MIME multipart - MIME Multipart Media Encapsulation	<=	
⊞ MIOP - Unreliable Multicast Inter-ORB Protocol	contains	
⊞ MIPv6 - Mobile IPv6 / Network Mobility	matches	
⊞ MMS - MMS		
⊞ MMSE - MMS Message Encapsulation		
⊞ Mobile IP - Mobile IP		
⊞ Modbus/TCP - Modbus/TCP		

Range (offset:length)

OK Cancel

Filtering Packets (3/4)

Examples:

- Capture only traffic to or from IP address 172.18.5.4
  - **host 172.18.5.4**
- Capture traffic to or from a range of IP addresses
  - **net 192.168.0.0/24**
  - **net 192.168.0.0 mask 255.255.255.0**
- Capture traffic from a range of IP addresses
  - **src net 192.168.0.0/24**
  - **src net 192.168.0.0 mask 255.255.255.0**
- Capture traffic to a range of IP addresses
  - **dst net 192.168.0.0/24**
  - **dst net 192.168.0.0 mask 255.255.255.0**
- Capture only DNS (port 53) traffic
  - **port 53**
- Capture non-HTTP and non-SMTP traffic on your server
  - **host www.example.com and not (port 80 or port 25)**
  - **host www.example.com and not port 80 and not port 25**

Filtering Packets (4/4)

Examples:

- Capture except all ARP and DNS traffic
  - **port not 53 and not arp**
- Capture traffic within a range of ports
  - **(tcp[2:2] > 1500 and tcp[2:2] < 1550) or (tcp[4:2] > 1500 and tcp[4:2] < 1550)**
  - **tcp portrange 1501-1549**
- Capture only Ethernet type EAPOL
  - **ether proto 0x888e**
- Capture only IP traffic

  (the shortest filter, but sometimes very useful to get rid of lower layer protocols like ARP and STP)
  - **ip**
- Capture only unicast traffic

  (useful to get rid of noise on the network if you only want to see traffic to and from your machine, not, for example, broadcast and multicast announcements)
  - **not broadcast and not multicast**

Protocol Hierarchy

Tucker Ellis & West Obsolete_Packets.cap - Wireshark

File Edit View Go Capture Analyze Statistics Help

Summary
Protocol Hierarchy
Conversations
Endpoints
IO Graphs
Conversation List
Endpoint List
Service Response Time
ANSI
Fax T38 Analysis...
GSM
H.225...
MTP3
RTP
SCTP
SIP...
VoIP Calls
WAP-WSP...
BOOTP-DHCP...
Destinations...
Flow Graph...
HTTP
IP address...
ISUP Messages...
Multicast Streams
ONC-RPC Programs
Packet Length...
Port Type...
SMPP Operations...
TCP Stream Graph
WLAN Traffic...

Filter: Expression... Clear Apply

802.11 Channel: Decryption Mode: None

No.	Time	Source	Protocol	Info
1	0.000000	::	ICMPv6	Multicast listener repo
2	0.000010	::	ICMPv6	Multicast listener repo
3	2.179063	192.168.1	NBNS	Name query NB LOCALHOST
4	2.439522	192.168.1	NBNS	Name query NB LOCALHOST
5	2.715733	192.168.1	NBNS	Name query NB LOCALHOST
6	2.821401	192.168.1	DNS	Standard query PTR 66.1
7	2.821546	192.168.1	DNS	Standard query PTR 255.
8	2.824683	192.168.1	DNS	Standard query response
9	2.990859	192.168.1	NBNS	Name query NB LOCALHOST
10	3.266913	192.168.1	NBNS	Name query NB LOCALHOST
11	3.495707	fe80::20	ICMPv6	Router solicitation
12	3.495727	fe80::20	ICMPv6	Router solicitation
13	3.542893	192.168.1	DNS	Standard query A DoCoMo
14	3.543088	192.168.1	NBNS	Name query NB LOCALHOST

Frame 1 (88 bytes on wire,
Linux cooked capture
Internet Protocol Version
Internet Control Message Pr

File: "C:\Users\ro2.TEW\Downloads\Obsolete_Packets.cap... Packets: 10949 Displayed: 10949 Marked: 0 Profile: Default

Protocol Hierarchy

Wireshark: Protocol Hierarchy Statistics

Display filter: none

Protocol	% Packets	Packets	Bytes	Mbit/s	End Packets	End Bytes	End Mbit/s
Frame	100.00%	10949	1433310	0.004	0	0	0.000
Linux cooked-mode capture	100.00%	10949	1433310	0.004	0	0	0.000
Internet Protocol Version 6	0.16%	18	1392	0.000	0	0	0.000
Internet Control Message Protocol v6	0.16%	18	1392	0.000	18	1392	0.000
Internet Protocol	82.62%	9046	1312691	0.004	0	0	0.000
User Datagram Protocol	17.33%	1898	262866	0.001	0	0	0.000
Transmission Control Protocol	64.69%	7083	1046121	0.003	2350	163598	0.000
Internet Group Management Protocol	0.57%	62	3440	0.000	62	3440	0.000
Internet Control Message Protocol	0.03%	3	264	0.000	3	264	0.000
DEC DNA Routing Protocol	2.60%	285	14820	0.000	285	14820	0.000
Address Resolution Protocol	7.63%	835	46928	0.000	835	46928	0.000
MS Network Load Balancing	1.26%	138	8280	0.000	138	8280	0.000
Data	2.75%	301	25143	0.000	301	25143	0.000
Logical-Link Control	2.23%	244	20024	0.000	0	0	0.000
Appletalk Address Resolution Protocol	0.37%	40	2480	0.000	40	2480	0.000
Internetwork Packet eXchange	1.46%	160	14328	0.000	0	0	0.000
Datagram Delivery Protocol	0.40%	44	3216	0.000	0	0	0.000
Internetwork Packet eXchange	0.27%	30	1680	0.000	0	0	0.000
Banyan Vines IP	0.47%	52	2352	0.000	0	0	0.000

Help Close

Follow TCP Stream

Tucker Ellis & West http-ethereal-trace-1 - Wireshark

File Edit View Go Capture Analyze Statistics Help

Filter: (ip.addr eq 192.168.1.102 and ip.addr eq 128.1 Expression... Clear Apply

802.11 Channel: Channel Offset: FCS Filter: Decryption Mode: None

No.	Time	Source	Destination	Protocol	Info
7	4.675312	192.168.1.102	128.119.245.12	TCP	unikeypro > http [SYN]
8	4.694429	128.119.245.12	192.168.1.1		> unikeypro [SYN,
9	4.694458	192.168.1.102	128.119.245		ypro > http [ACK]
10	4.694850	192.168.1.102	128.119.245		ethereal-labs/lab
11	4.717289	128.119.245.12	192.168.1.1		> unikeypro [ACK]
12	4.718993	128.119.245.12	192.168.1.1		1.1 200 OK (text/
13	4.724332	192.168.1.102	128.119.245		favicon.ico HTTP/1
14	4.750366	128.119.245.12	192.168.1.1		1.1 404 Not Found
15	4.859777	192.168.1.102	128.119.245		ypro > http [ACK]

Mark Packet (toggle)
Set Time Reference (toggle)
Apply as Filter
Prepare a Filter
Conversation Filter
Colorize Conversation
SCTP
Follow TCP Stream
Follow UDP Stream
Follow SSL Stream
Copy
Export Selected Packet Bytes...
Decode As...
Print...
Show Packet in New Window

Frame 8 (62 bytes on wire, 62 bytes captured)
Ethernet II, Src: LinksysG_da:af:73 (00:06:25:da:af 5:23 (00:08:74:4f:36
Internet Protocol, Src: 128.119.245.12 (128.119.245 (192.168.1.102)
Transmission Control Protocol, Src Port: http (80), 27), Seq: 0, Ack: 1,

```
0000  00 08 74 4f 36 23 00 06  25 da af 73 08 00 45 00   ..tO6#.. %..s..E.
0010  00 30 00 00 40 00 37 06  0c 36 80 77 f5 0c c0 a8   .0..@.7. .6.w....
0020  01 66 00 50 10 1f 6b a6  54 91 f5 32 64 b2 70 12   .f.P..k. T..2d.p.
0030  16 d0 0a 21 00 00 02 04  05 b4 01 01 04 02         ...!.... ......
```

File: "C:\Traces\http-ethereal-trace-1" 4443 Bytes 00:00:06 Packets: 1... Profile: Def...

Follow TCP Stream

red - stuff you sent **blue - stuff you get**

Follow TCP Stream

Stream Content

```
GET /ethereal-labs/lab2-1.html HTTP/1.1
Host: gaia.cs.umass.edu
User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120
Netscape/7.01
Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/
plain;q=0.8,video/x-mng,image/png,image/jpeg,image/gif;q=0.2,text/css,*/*;q=0.1
Accept-Language: en-us, en;q=0.50
Accept-Encoding: gzip, deflate, compress;q=0.9
Accept-Charset: ISO-8859-1, utf-8;q=0.66, *;q=0.66
Keep-Alive: 300
Connection: keep-alive

HTTP/1.1 200 OK
Date: Tue, 23 Sep 2003 05:29:50 GMT
Server: Apache/2.0.40 (Red Hat Linux)
Last-Modified: Tue, 23 Sep 2003 05:29:00 GMT
ETag: "1bfed-49-79d5bf00"
Accept-Ranges: bytes
Content-Length: 73
Keep-Alive: timeout=10, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=ISO-8859-1

<html>
Congratulations.  You've downloaded the file lab2-1.html!
</html>
GET /favicon.ico HTTP/1.1
Host: gaia.cs.umass.edu
User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120
Netscape/7.01
Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/
plain;q=0.8,video/x-mng,image/png,image/jpeg,image/gif;q=0.2,text/css,*/*;q=0.1
Accept-Language: en-us, en;q=0.50
```

Find | Save As | Print | Entire conversation (2714 bytes) | ASCII | EBCDIC | Hex Dump | C Arrays | Raw

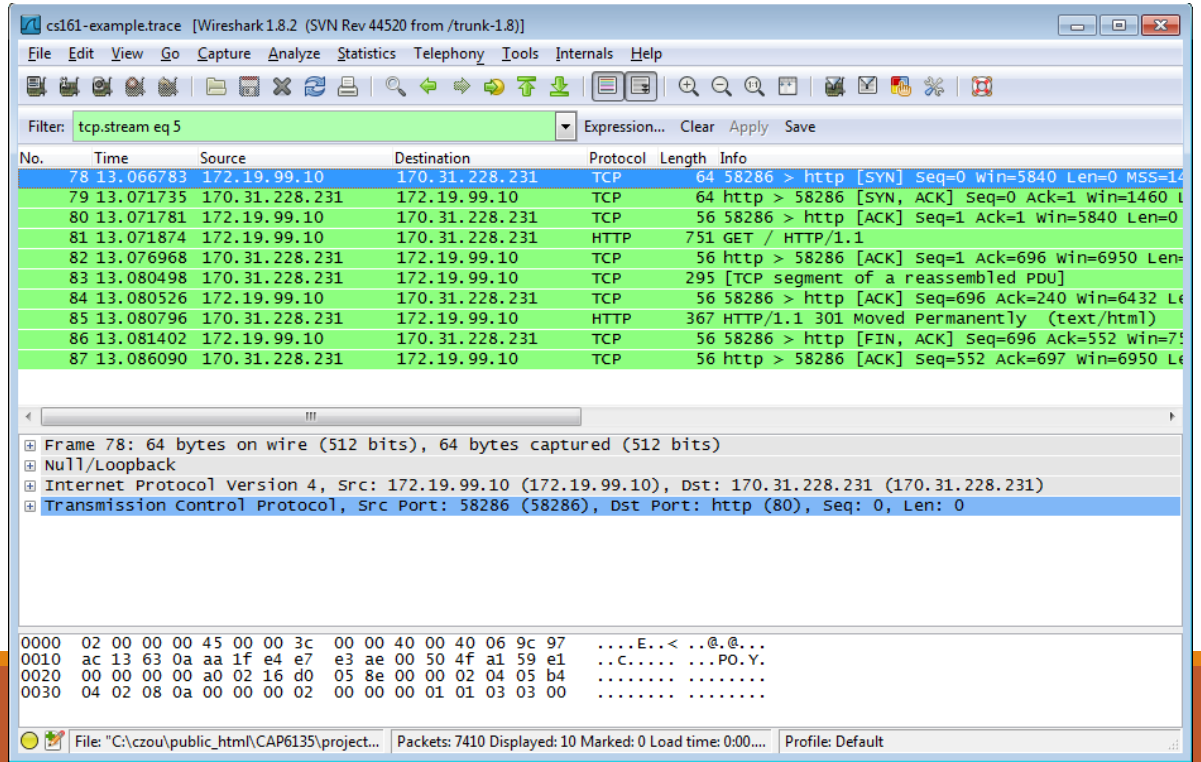
Help | Close | Filter Out This Stream

Filter out/in Single TCP Stream

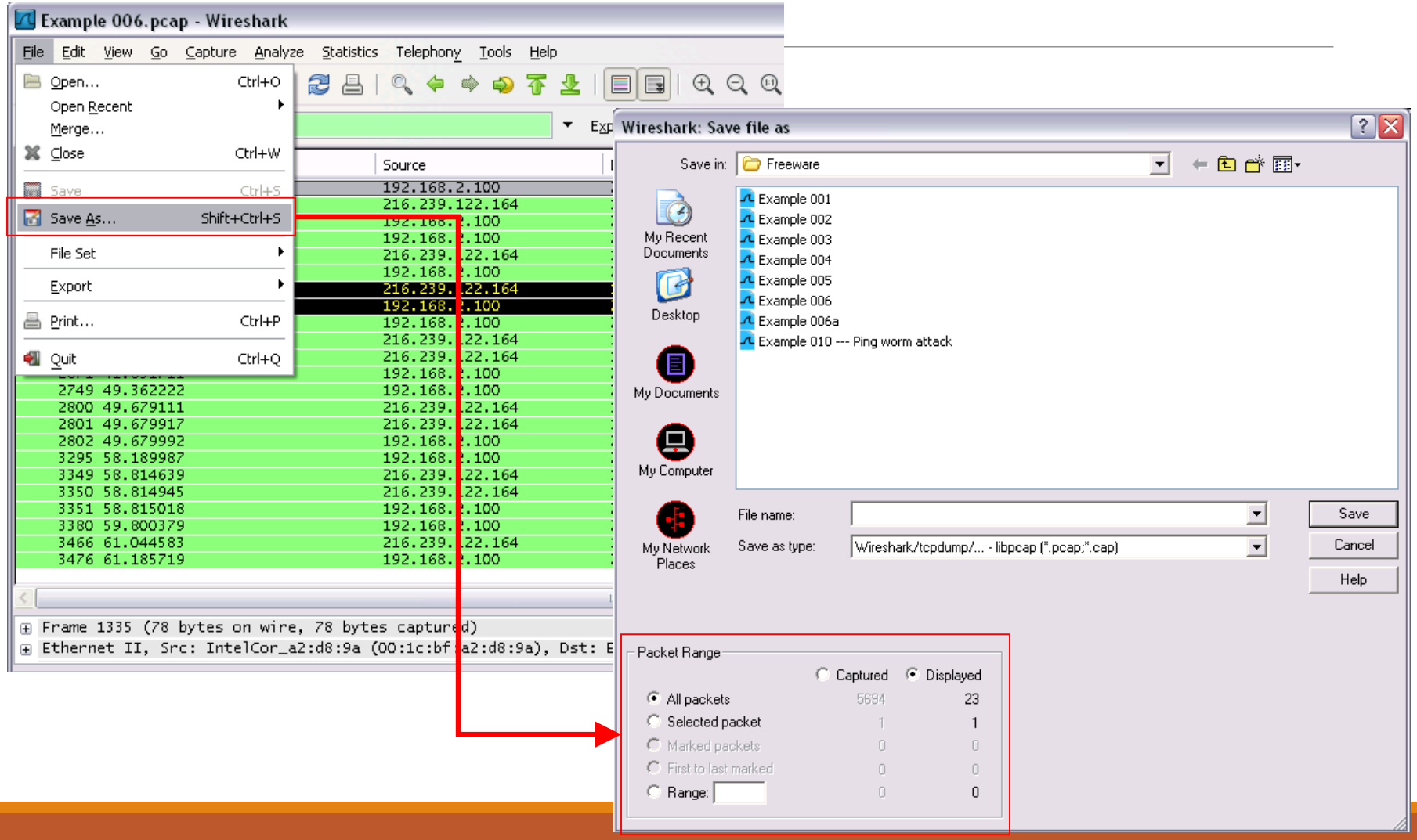
When click “filter out this TCP stream” in previous page’s box, new filter string will contain like:

- http and !(tcp.stream eq 5)

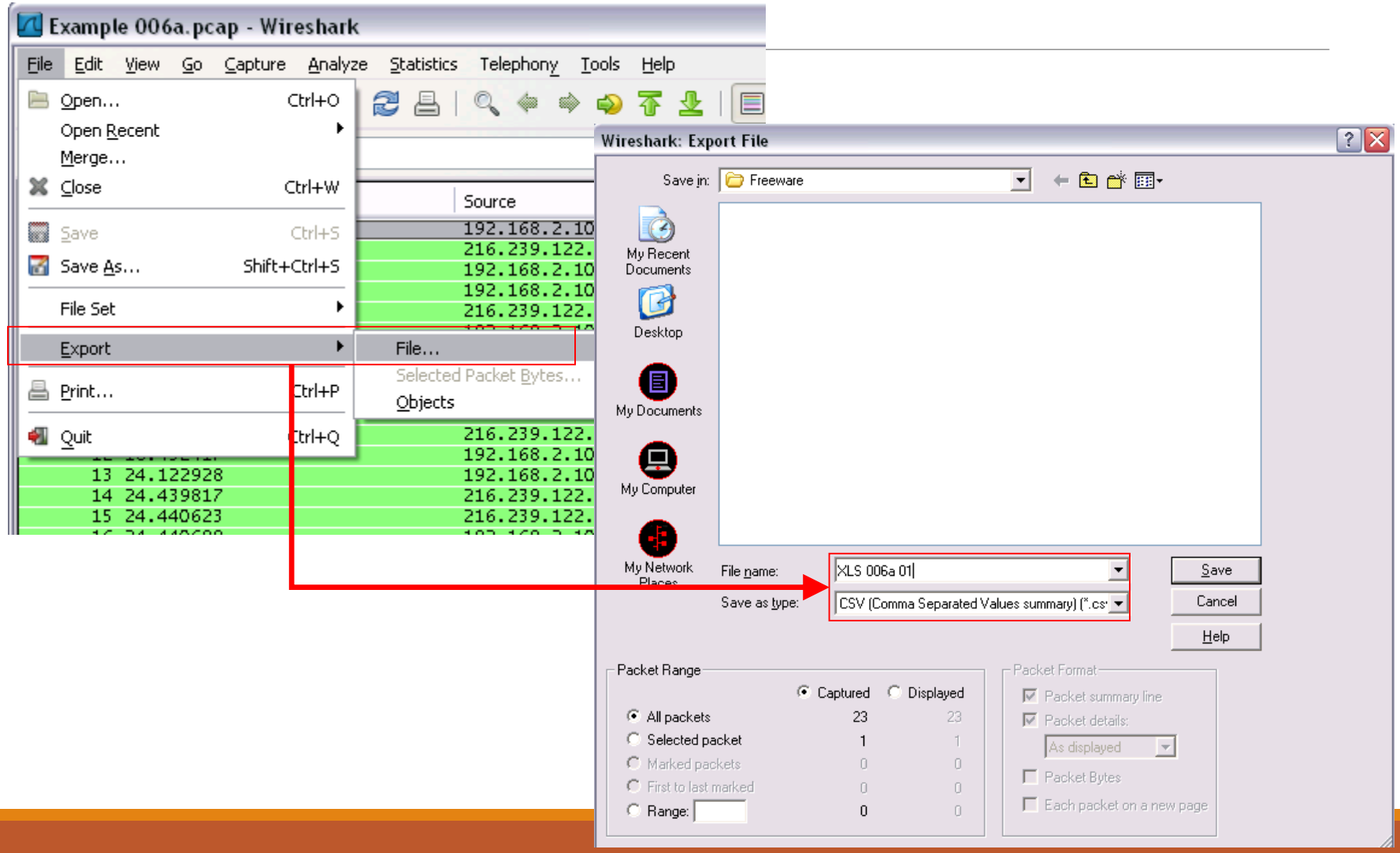
So, if you use “tcp.stream eq 5” as filter string, you keep this HTTP session



Saving and Manipulating Packets (1/3)



Saving and Manipulating Packets (2/3)



Saving and Manipulating Packets (3/3)

No.	Time	Time Variation	Source	Destination	Protocol	Info
1	0	0	192.168.2.100	216.239.122.164	TCP	27837 > http [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=1 TSV=0 TSER=0
2	0.226724	0.226724	216.239.122.164	192.168.2.100	TCP	http > 27837 [SYN, ACK] Seq=0 Ack=1 Win=8190 Len=0 MSS=1380
3	0.226772	4.8E-05	192.168.2.100	216.239.122.164	TCP	27837 > http [ACK] Seq=1 Ack=1 Win=65535 Len=0
4	0.227146	0.227098	192.168.2.100	216.239.122.164	HTTP	GET /i/b.jpg HTTP/1.1
5	0.700674	0.473576	216.239.122.164	192.168.2.100	HTTP	HTTP/1.1 200 OK (JPEG JFIF image)
6	0.883533	0.409957	192.168.2.100	216.239.122.164	TCP	27837 > http [ACK] Seq=649 Ack=767 Win=64769 Len=0
7	**1.161312**	**0.751355**	**216.239.122.164**	**192.168.2.100**	**HTTP**	**[TCP Retransmission] HTTP/1.1 200 OK (JPEG JFIF image)**
8	1.161361	0.410006	192.168.2.100	216.239.122.164	TCP	[TCP Dup ACK 6#1] 27837 > http [ACK] Seq=649 Ack=767 Win=64769 Len=0
9	16.211468	15.801462	192.168.2.100	216.239.122.164	HTTP	GET /i/b.jpg HTTP/1.1
10	16.452024	0.650562	216.239.122.164	192.168.2.100	TCP	[TCP segment of a reassembled PDU]
11	16.452343	15.801781	216.239.122.164	192.168.2.100	HTTP	HTTP/1.1 200 OK (JPEG JFIF image)
12	16.452417	0.650636	192.168.2.100	216.239.122.164	TCP	27837 > http [ACK] Seq=1539 Ack=1533 Win=65535 Len=0
13	24.122928	23.472292	192.168.2.100	216.239.122.164	HTTP	GET /i/b.jpg HTTP/1.1
14	24.439817	0.967525	216.239.122.164	192.168.2.100	TCP	[TCP segment of a reassembled PDU]
15	24.440623	23.473098	216.239.122.164	192.168.2.100	HTTP	HTTP/1.1 200 OK (JPEG JFIF image)
16	24.440698	0.9676	192.168.2.100	216.239.122.164	TCP	27837 > http [ACK] Seq=2384 Ack=2299 Win=64769 Len=0
17	32.950693	31.983093	192.168.2.100	216.239.122.164	HTTP	GET /i/b.jpg HTTP/1.1
18	33.575345	1.592252	216.239.122.164	192.168.2.100	TCP	[TCP segment of a reassembled PDU]
19	33.575651	31.983399	216.239.122.164	192.168.2.100	HTTP	HTTP/1.1 200 OK (JPEG JFIF image)
20	33.575724	1.592325	192.168.2.100	216.239.122.164	TCP	27837 > http [ACK] Seq=3269 Ack=3065 Win=65535 Len=0
21	34.561085	32.96876	192.168.2.100	216.239.122.164	HTTP	GET /b.gif HTTP/1.1
22	35.805289	2.836529	216.239.122.164	192.168.2.100	HTTP	HTTP/1.1 200 OK (GIF89a)
23	35.946425	33.109896	192.168.2.100	216.239.122.164	TCP	27837 > http [ACK] Seq=4080 Ack=3567 Win=65033 Len=0

Packet Statistics (1/8)

P

Wireshark: Protocol Hierarchy Statistics

Display filter: none

Protocol	% Packets	Packets	Bytes	Mbit/s	End Packets	End Bytes	End Mbit/s
Frame	100.00 %	1276	385508	0.030	0	0	0.000
Ethernet	100.00 %	1276	385508	0.030	0	0	0.000
Internet Protocol	99.69 %	1272	385340	0.030	0	0	0.000
Internet Control Message Protocol	8.23 %	105	7770	0.001	105	7770	0.001
User Datagram Protocol	4.86 %	62	11029	0.001	0	0	0.000
Simple Network Management Protocol	1.57 %	20	1825	0.000	20	1825	0.000
Bootstrap Protocol	0.31 %	4	1864	0.000	4	1864	0.000
NetBIOS Name Service	0.24 %	3	276	0.000	3	276	0.000
Domain Name Service	2.66 %	34	6922	0.001	34	6922	0.001
Data	0.08 %	1	142	0.000	1	142	0.000
Transmission Control Protocol	86.60 %	1105	366541	0.028	658	149347	0.011
Post Office Protocol	1.41 %	18	1486	0.000	18	1486	0.000
Hypertext Transfer Protocol	33.62 %	429	215708	0.017	388	188597	0.014
Line-based text data	2.27 %	29	21877	0.002	29	21877	0.002
JPEG File Interchange Format	0.08 %	1	59	0.000	1	59	0.000
eXtensible Markup Language	0.39 %	5	2730	0.000	5	2730	0.000
Media Type	0.16 %	2	1160	0.000	2	1160	0.000
Compuserve GIF	0.31 %	4	1285	0.000	4	1285	0.000
Address Resolution Protocol	0.31 %	4	168	0.000	4	168	0.000

Help Close

Packet Statistics (2/8)

Conversations: (Untitled)

Ethernet: 4 | Fibre Channel | FDDI | IPv4: 38 | IPX | JXTA | NCP | RSVP | SCTP | TCP: 79 | Token Ring | UDP: 23 | USB | WLAN

IPv4 Conversations

Address A	Address B	Packets	Bytes	Packets A->B	Bytes A->B	Packets A<-B	Bytes A<-B	Rel Start	Duratior
192.168.2.100	255.255.255.255	1	142	1	142	0	0	96.384838000	0.0000
192.168.2.101	255.255.255.255	2	684	2	684	0	0	47.852757000	3.0721
192.168.2.1	255.255.255.255	2	1180	2	1180	0	0	47.857905000	3.0722
192.168.2.100	212.143.162.144	10	2194	6	815	4	1379	87.473054000	65.0878
192.168.2.100	212.179.31.90	10	1342	6	971	4	371	91.655266000	60.9113
62.90.102.31	192.168.2.100	10	1373	4	441	6	932	91.660203000	60.9174
192.168.2.100	212.150.22.226	10	1327	6	956	4	371	91.732692000	60.8200
192.168.2.100	212.150.236.220	10	1470	6	981	4	489	91.742363000	60.8122
192.168.2.100	212.179.58.84	10	1725	6	954	4	771	92.287214000	2.4984
82.80.238.109	192.168.2.100	11	1282	5	440	6	842	91.646648000	60.9214
10.12.44.2	192.168.2.100	12	888	6	444	6	444	31.421091000	164.384
10.10.10.2	192.168.2.100	12	888	6	444	6	444	31.531676000	164.804
10.12.20.2	192.168.2.100	12	888	6	444	6	444	5.250457000	164.523
10.31.68.1	192.168.2.100	12	888	6	444	6	444	5.578447000	164.631
10.100.102.2	192.168.2.100	12	888	6	444	6	444	17.858674000	164.363

Name resolution

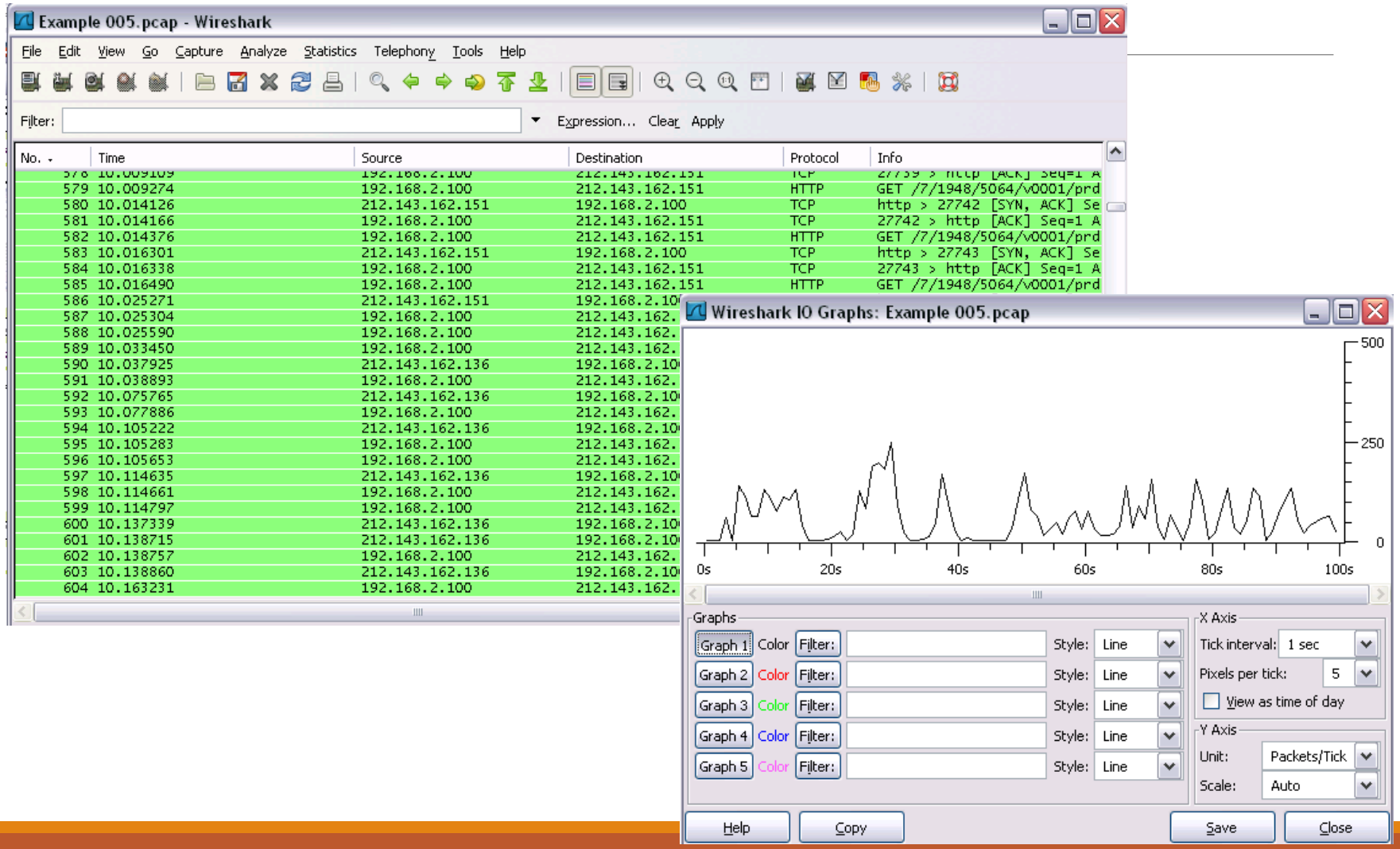
Limit to display filter

Help | Copy | Close

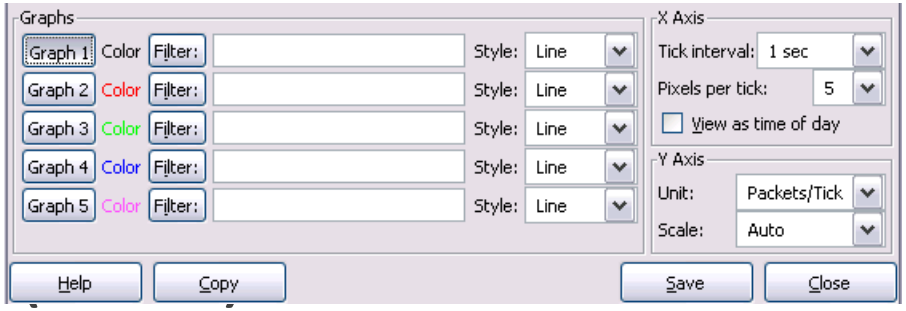
With some manipulation

	A	B	C	D	E	F	G	H	I	J	K	L
1	Address A	Address B	Packets	Bytes	Packets A->B	Bytes A->B	Packets A<-B	Bytes A<-B	Rel Start	Duration	bps A->B	bps A<-B
2	10.10.10.1	10.159.3.103	2	124	0	0	2	124	0	42.0039	N/A	23.62
3	1.1.1.1	10.159.3.103	2	120	0	0	2	120	24.49414	1.1885	N/A	807.76
4	1.1.1.1	127.0.0.1	4	248	4	248	0	0	10.713867	14.9814	132.43	N/A
5	10.159.3.103	212.179.1.202	491	458158	185	10643	306	447515	23.216796	15.4082	5525.89	232351.54

Packet Statistics (3/8)



Packet Statistics



Configurable Options

- I/O Graphs
  - **Graph 1-5**: enable the specific graph 1-5 (graph 1 by default)
  - **Filter**: a display filter for this graph (only the packets that pass this filter will be taken into account for this graph)
  - **Style**: the style of the graph (Line/Impulse/FBar/Dot)
- X Axis
  - **Tick interval**: an interval in x direction lasts (10/1 minutes or 10/1/0.1/0.01/0.001 seconds)
  - **Pixels per tick**: use 10/5/2/1 pixels per tick interval
  - **View as time of day**: option to view x direction labels as time of day instead of seconds or minutes since beginning of capture
- Y Axis
  - **Unit**: the unit for the y direction (Packets/Tick, Bytes/Tick, Bits/Tick, Advanced...)
  - **Scale**: the scale for the y unit (Logarithmic,Auto,10,20,50,100,200,...)

Packet Statistics (5/8)

Snif1 --- File copy from other side.cap - Wireshark

File Edit View Go Capture Analyze Statistics Telephony Tools Help

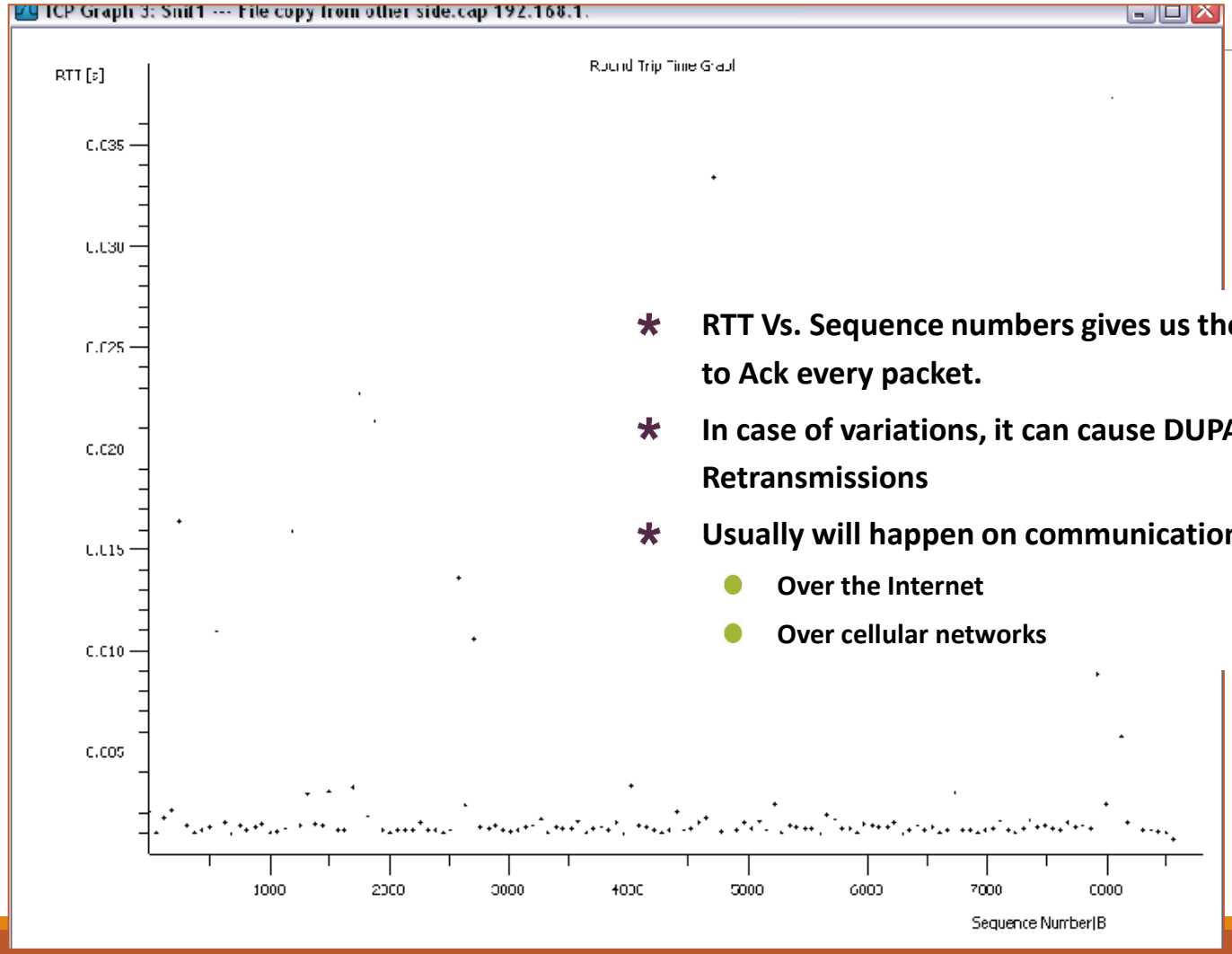
Filter: (tcp.stream eq 0) Expression... Clear Apply

- Summary
- Protocol Hierarchy
- Conversations
- Endpoints
- Packet Lengths...
- IO Graphs
- Conversation List
- Endpoint List
- Service Response Time
- BOOTP-DHCP...
- Compare...
- Flow Graph...
- HTTP
- IP Addresses...
- IP Destinations...
- IP Protocol Types...
- ONC-RPC Programs
- TCP Stream Graph
  - Round Trip Time Graph
  - Throughput Graph
  - Time-Sequence Graph (Stevens)
  - Time-Sequence Graph (tcptrace)
- UDP Multicast Streams
- WLAN Traffic...

No.	Time		Destination	Protocol	Info
762	0.132867		192.168.1.102	TCP	[TCP segment of a reassembled PDU]
763	0.132992		192.168.1.102	TCP	[TCP segment of a reassembled PDU]
764	0.133116		192.168.1.102	TCP	[TCP segment of a reassembled PDU]
766	0.133247		192.168.1.102	TCP	[TCP segment of a reassembled PDU]
767	0.133258		192.168.1.102	SMB	Read AndX Response, 61440 bytes
768	0.133265		192.168.104.77	TCP	paradym-31port > microsoft-ds [ACK] Seq=883 Ack=686944 Win
769	0.133272		192.168.104.77	TCP	paradym-31port > microsoft-ds [ACK] Seq=883 Ack=689864 Win
770	0.133281		192.168.104.77	TCP	paradym-31port > microsoft-ds [ACK] Seq=883 Ack=692784 Win
771	0.134377		192.168.104.77	TCP	paradym-31port > microsoft-ds [ACK] Seq=883 Ack=695704 Win
773	0.139299		192.168.104.77	SMB	Read AndX Request, FID: 0x8003, 61440 bytes at offset 1747
774	0.140539		192.168.1.102	TCP	[TCP segment of a reassembled PDU]
775	0.140661		192.168.1.102	TCP	[TCP segment of a reassembled PDU]
776	0.140785		192.168.1.102	TCP	[TCP segment of a reassembled PDU]
777	0.140909		192.168.1.102	TCP	[TCP segment of a reassembled PDU]
778	0.141033		192.168.1.102	TCP	[TCP segment of a reassembled PDU]
779	0.141155		192.168.1.102	TCP	[TCP segment of a reassembled PDU]
780	0.141282		192.168.1.102	TCP	[TCP segment of a reassembled PDU]
781	0.141293		192.168.104.77	TCP	paradym-31port > microsoft-ds [ACK] Seq=946 Ack=698807 Win
782	0.141411				[TCP segment of a reassembled PDU]
783	0.141421				paradym-31port > microsoft-ds [ACK] Seq=946 Ack=701727 Win
784	0.141538				[TCP segment of a reassembled PDU]
785	0.141663				[TCP segment of a reassembled PDU]
786	0.141783				[TCP segment of a reassembled PDU]
787	0.141908	192.168.104.77			[TCP segment of a reassembled PDU]
788	0.142035	192.168.104.77	192.168.1.102	TCP	[TCP segment of a reassembled PDU]
789	0.142046	192.168.1.102	192.168.104.77	TCP	paradym-31port > microsoft-ds [ACK] Seq=946 Ack=704647 Win
790	0.142053	192.168.1.102	192.168.104.77	TCP	paradym-31port > microsoft-ds [ACK] Seq=946 Ack=707567 Win

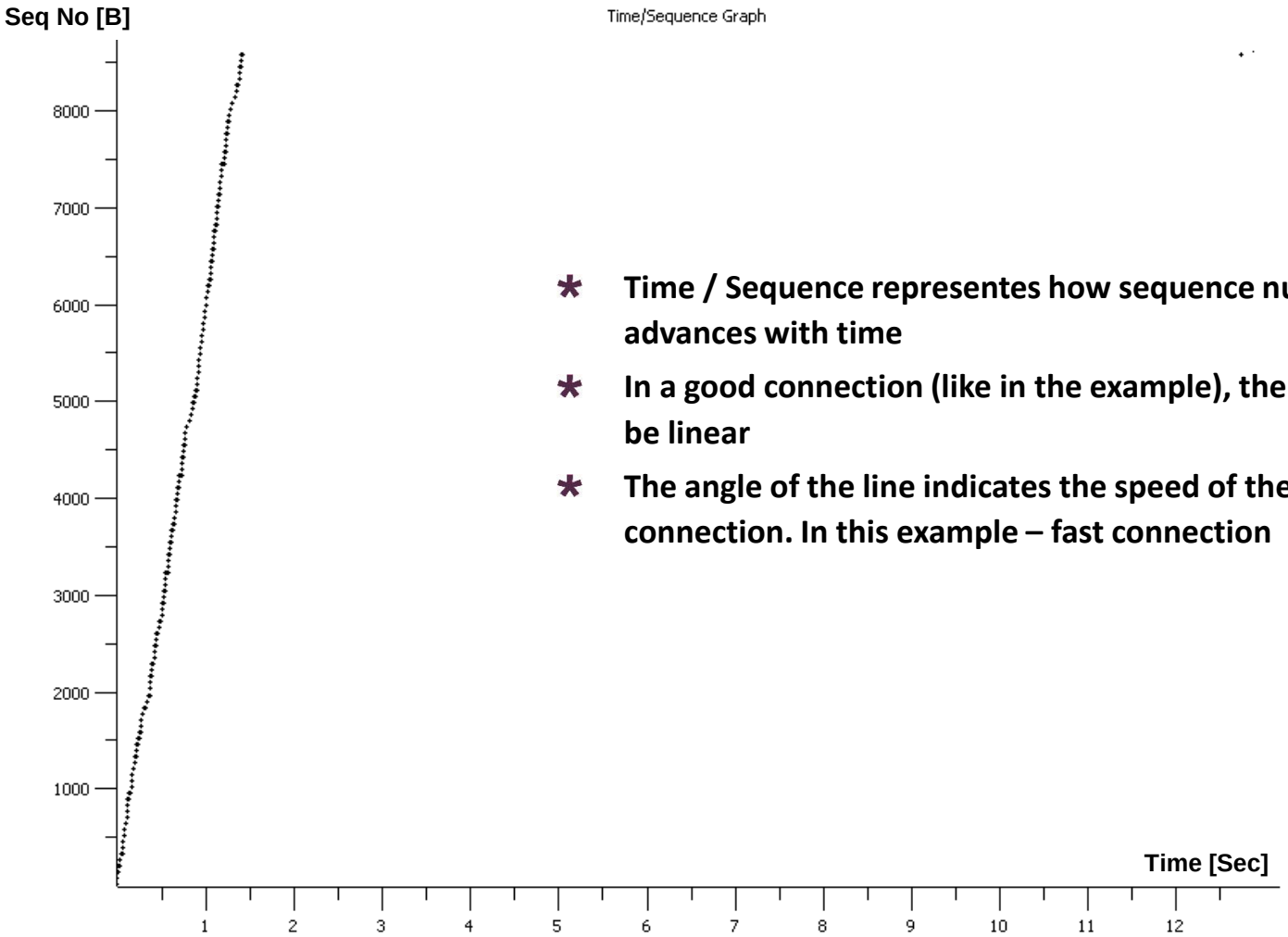
```
Frame 771 (60 bytes on wire, 60 bytes captured)
Ethernet II, Src: Intel_4c:cc:89 (00:d0:b7:4c:cc:89), Dst: Intel_2e:32:a9 (00:90:27:2e:32:a9)
Internet Protocol, Src: 192.168.1.102 (192.168.1.102), Dst: 192.168.104.77 (192.168.104.77)
Transmission Control Protocol, Src Port: paradym-31port (1864), Dst Port: microsoft-ds (445), Seq: 883, Ack: 695704, Len: 0
```

Packet Statistics (6/8)



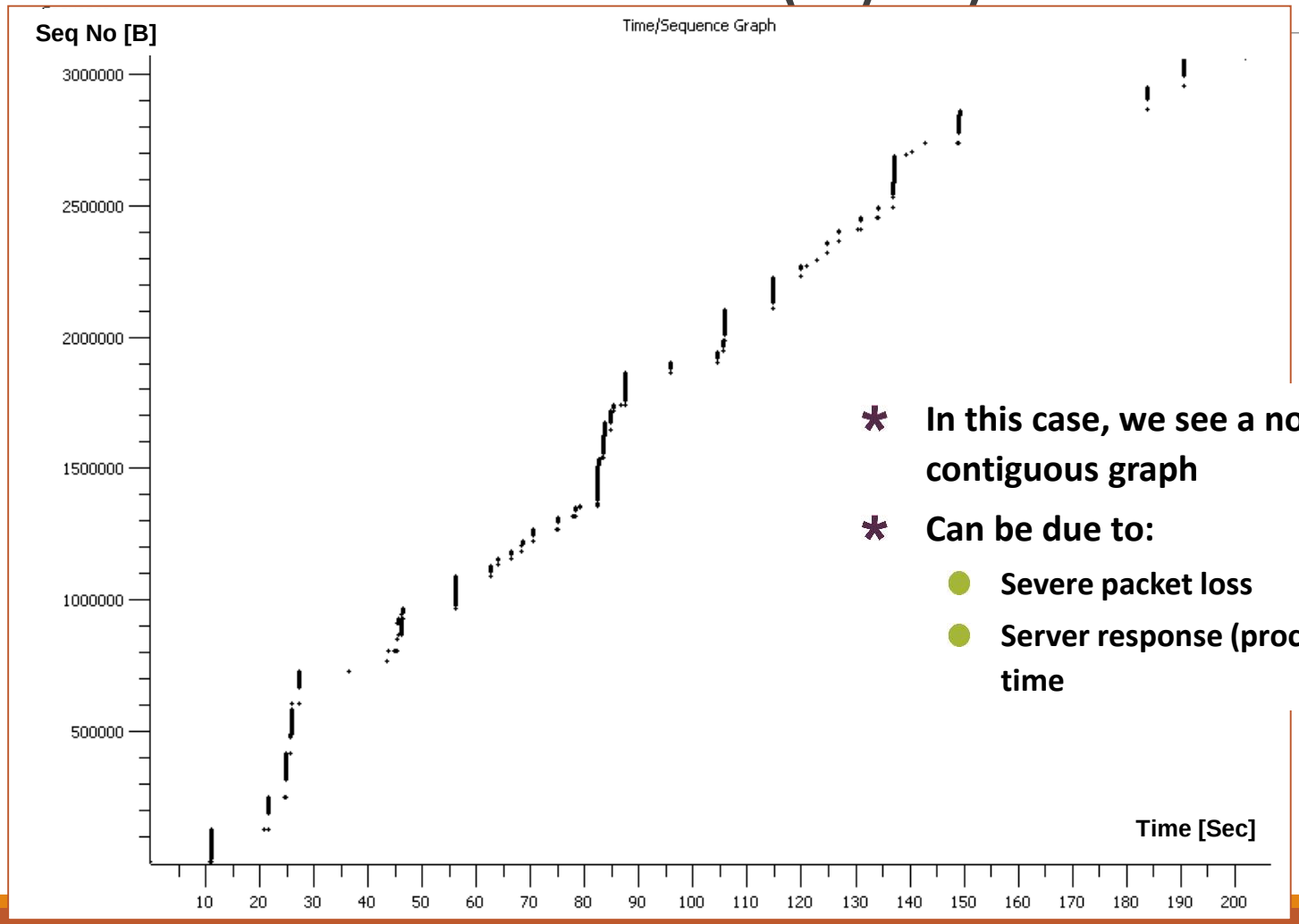
- **RTT Vs. Sequence numbers gives us the time that take to Ack every packet.**
- **In case of variations, it can cause DUPACKs and even Retransmissions**
- **Usually will happen on communications lines:**
  - **Over the Internet**
  - **Over cellular networks**

Packet Statistics (7/8)



- **Time / Sequence representes how sequence numbers advances with time**
- **In a good connection (like in the example), the line will be linear**
- **The angle of the line indicates the speed of the connection. In this example – fast connection**

Packet Statistics (8/8)



- In this case, we see a non-contiguous graph
- Can be due to:
  - Severe packet loss
  - Server response (processing) time

Colorizing Specific Packets (1/4)

Packet Colorization

- Colorize packets according to a filter
- Allow to emphasize the packets interested in
- A lot of Coloring Rule examples at the Wireshark Wiki Coloring Rules page at http://wiki.wireshark.org/ColoringRules

We want to watch a specific protocol through out the capture file

No.	Time	Source	Destination	Protocol	Info
1	0.000000	172.16.2.255	172.17.220.62	ICMP	Echo (ping) request
2	0.001075	172.16.3.14	172.16.1.224	DCERPC	Request: call_id: 395 opnum: 2 ctx_id: 0
3	0.002838	172.16.1.224	172.16.3.14	DCERPC	Response: call_id: 395 ctx_id: 0
4	0.004758	172.16.1.20	172.16.2.236	HTTP	Continuation or non-HTTP traffic
5	0.005001	172.16.2.236	172.16.1.20	TCP	netview-aix-12 > http-alt [ACK] Seq=1 Ack=1461 Win=64240 Len=0
6	0.005134	172.16.1.20	172.16.2.236	HTTP	Continuation or non-HTTP traffic
7	0.005658	172.16.1.20	172.16.2.236	HTTP	Continuation or non-HTTP traffic
8	0.005790	172.16.1.20	172.16.2.236	HTTP	Continuation or non-HTTP traffic
9	0.005906	172.16.2.236	172.16.1.20	TCP	netview-aix-11 > http-alt [ACK] Seq=1 Ack=2049 Win=64240 Len=0
10	0.006231	172.16.1.20	172.16.2.236	HTTP	Continuation or non-HTTP traffic
11	0.006253	172.16.1.20	172.16.2.236	HTTP	Continuation or non-HTTP traffic
12	0.006444	172.16.2.236	172.16.1.20	TCP	netview-aix-11 > http-alt [ACK] Seq=1 Ack=4097 Win=64240 Len=0
13	0.006826	172.16.1.20	172.16.2.236	HTTP	Continuation or non-HTTP traffic
14	0.006962	172.16.1.20	172.16.2.236	HTTP	Continuation or non-HTTP traffic
15	0.007079	172.16.2.236	172.16.1.20	TCP	netview-aix-11 > http-alt [ACK] Seq=1 Ack=6145 Win=64240 Len=0
16	0.007221	172.16.3.129	172.16.201.60	ICMP	Echo (ping) request
17	0.007951	172.16.1.20	172.16.2.236	HTTP	Continuation or non-HTTP traffic
18	0.007972	64.236.34.97	172.16.2.219	TCP	http > metasage [ACK] Seq=1 Ack=1 Win=4096 Len=0
19	0.008068	172.16.1.20	172.16.2.236	HTTP	Continuation or non-HTTP traffic
20	0.008263	172.16.1.20	172.16.2.236	HTTP	Continuation or non-HTTP traffic
21	0.008279	172.16.1.40	172.16.2.5	TCP	av-us > radmin-port [PSH, ACK] Seq=1 Ack=1 Win=16306 Len=14

File Edit View Go Capture Analyze Statistics Telephony Tools Help

Filter: Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
1	0.000000	172.16.2.255	172.17.220.62	ICMP	Echo (ping) request
2	0.001075	172.16.3.14	172.16.1.224	DCERPC	Request: call_id: 395 opnum: 2 ctx_id: 0
3	0.002838	172.16.1.224	172.16.3.14	DCERPC	tx_id: 0
4	0.004758	172.16.1.20	172.16.2.236	HTTP	traffic
5	0.005001	172.16.2.236	172.16.1.20	TCP	t [ACK] Seq=1 Ack=1461 Win=64240 Len=0
6	0.005134	172.16.1.20	172.16.2.236	HTTP	traffic
7	0.005658	172.16.1.20	172.16.2.236	HTTP	traffic
8	0.005790	172.16.1.20	172.16.2.236	HTTP	traffic
9	0.005906	172.16.2.236	172.16.1.20	TCP	t [ACK] Seq=1 Ack=2049 Win=64240 Len=0
10	0.006231	172.16.1.20	172.16.2.236	HTTP	
11	0.006253	172.16.1.20	172.16.2.236	HTTP	
12	0.006444	172.16.2.236	172.16.1.20	TCP	
13	0.006826	172.16.1.20	172.16.2.236	HTTP	
14	0.006962	172.16.1.20	172.16.2.236	HTTP	
15	0.007079	172.16.2.236	172.16.1.20	TCP	
16	0.007221	172.16.3.129	172.16.201.60	ICMP	
17	0.007951	172.16.1.20	172.16.2.236	HTTP	traffic
18	0.007972	64.236.34.97	172.16.2.219	TCP	q=1 Ack=1 Win=4096
19	0.008068	172.16.1.20	172.16.2.236	HTTP	traffic
20	0.008263	172.16.1.20	172.16.2.236	HTTP	traffic
21	0.008279	172.16.1.40	172.16.2.5	TCP	ACK] Seq=1 Ack=1

Mark Packet (toggle)
Set Time Reference (toggle)
Apply as Filter
Prepare a Filter
Conversation Filter
Colorize Conversation
SCTP
Follow TCP Stream
Follow UDP Stream
Follow SSL Stream
Copy
Decode As...
Print...
Show Packet in New Window

Ethernet
IP
TCP
UDP
PN-CBA Server

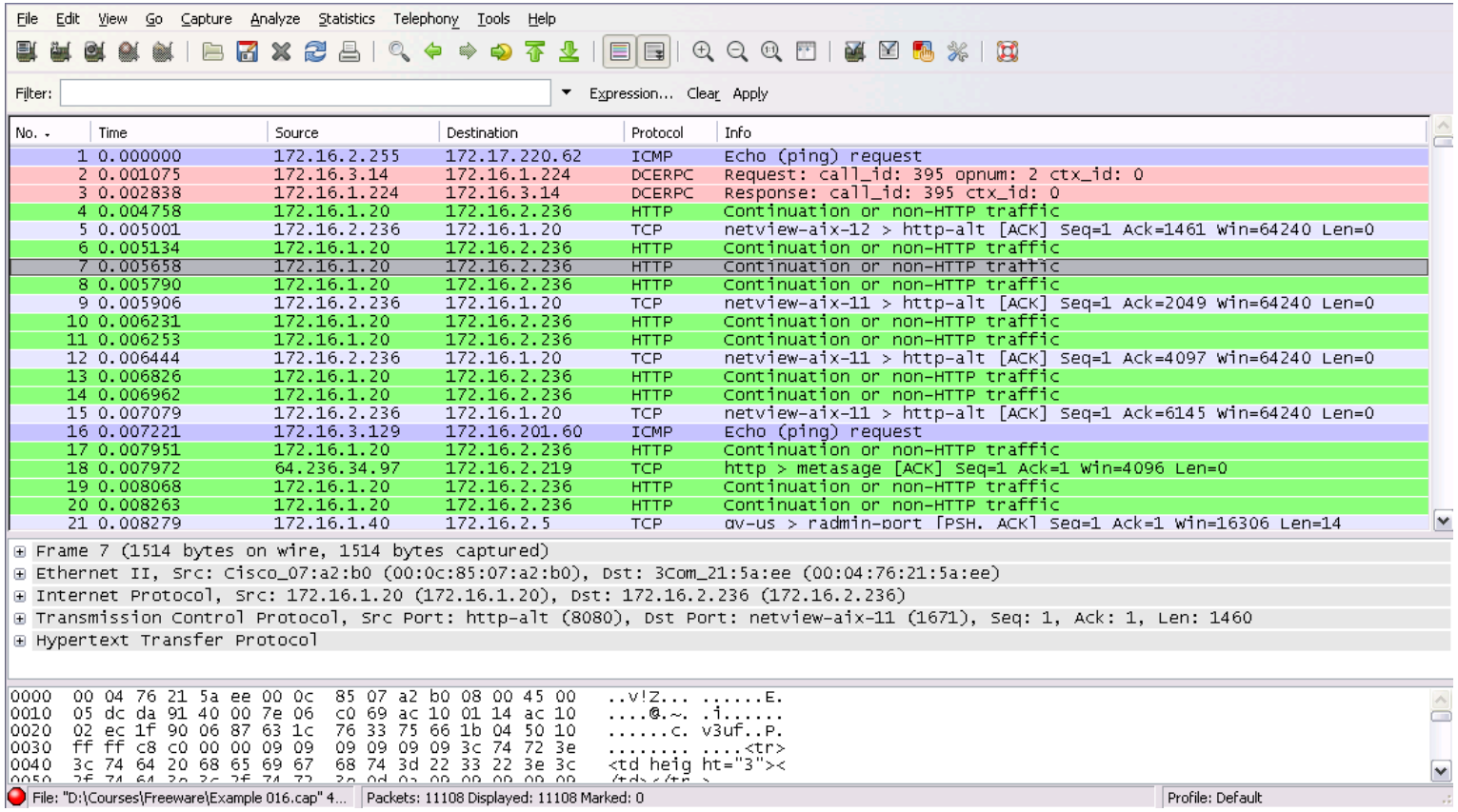
Color 1
Color 2
Color 3
Color 4
Color 5
Color 6
Color 7
Color 8
Color 9
Color 10
New Coloring Rule...

```
Frame 2 (198 bytes on wire, 198 bytes captured)
Ethernet II, Src: 3com_74:5a:2b (00:50:da:74:5a:2b), Dst: Cisco_07:a2:b0 (00:0c:85:07:a2:b0)
Internet Protocol, Src: 172.16.3.14 (172.16.3.14), Dst: 172.16.1.224 (172.16.1.224)
Transmission Control Protocol, Src Port: writesrv (1334), Dst Port: alta-ana-lm (1346), Seq: 1, Ack: 1, Len:
DCE RPC Request, Fragment: Single, FragLen: 144, Call: 395 Ctx: 0

0000  00 0c 85 07 a2 b0 00 50  da 74 5a 2b 08 00 45 00   .......P .tZ+..E.
0010  00 b8 58 74 40 00 80 06  44 bd ac 10 03 0e ac 10   ..Xt@... D.......
0020  01 e0 05 36 05 42 9a 8d  a4 a2 49 a0 ce 05 50 18   ...6.B.. ..I...P.
0030  fa 90 0f c3 00 00 05 00  00 03 10 00 00 00 90 00   ........ ........
0040  10 00 8b 01 00 00 52 00  00 00 00 00 02 00 00 00   ......R. ........
```

File: "D:\Customers\Mivtachim\Snif4 --- Center-... | Packets: 11108 Displayed: 11108 Marked: 0 | Profile: Default

Colorizing Specific Packets (3/4)



Colorizing Specific Packets (4/4)

Filter: Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
1	0.000000	192.168.2.100	198.65.166.131	UDP	Source port: 64064 Destination port: sip
2	1.708525	192.168.2.100	130.94.88.123	TCP	appworxsrv > https [FIN, ACK] Seq=1 Ack=1 Win=64240 Len=0
3	1.709469	192.168.2.100	130.94.88.123	TCP	lv-jc > https [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=1 TSV=0 TSER=
4	2.001023	130.94.88.123	192.168.2.100	TCP	https > lv-jc [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 WS=2
5	2.001077	192.168.2.100	130.94.88.123	TCP	lv-jc > https [ACK] Seq=1 Ack=1 Win=128480 Len=0
6	2.001180	130.94.88.123	192.168.2.100	TCP	https > appworxsrv [ACK] Seq=1 Ack=2 Win=3756 Len=0
7	2.001777	192.168.2.100	130.94.88.123	SSL	Client Hello
8	2.308152	130.94.88.123	192.168.2.100	TCP	https > lv-jc [ACK] Seq=1 Ack=103 Win=5840 Len=0
9	2.308490	130.94.88.123	192.168.2.100	TLSv1	Server Hello,
10	2.309543	130.94.88.123	192.168.2.100	TCP	[TCP segment of a reassembled PDU]
11	2.309618	192.168.2.100	130.94.88.123	TCP	lv-jc > https [ACK] Seq=103 Ack=2705 Win=128480 Len=0
12	2.617428	130.94.88.123	192.168.2.100	TCP	[TCP segment of a reassembled PDU]
13	2.619328	130.94.88.123	192.168.2.100	TLSv1	Certificate, Server Hello Done
14	2.619440	192.168.2.100	130.94.88.123	TCP	lv-jc > https [ACK] Seq=103 Ack=4549 Win=128480 Len=0
15	2.620478	192.168.2.100	130.94.88.123	TLSv1	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Messag
16	2.922741	130.94.88.123	192.168.2.100	TLSv1	Change Cipher Spec, Encrypted Handshake Message
17	2.926069	192.168.2.100	130.94.88.123	TCP	[TCP segment of a reassembled PDU]
18	2.926211	192.168.2.100	130.94.88.123	TLSv1	Application Data
19	3.229909	130.94.88.123	192.168.2.100	TCP	https > lv-jc [ACK] Seq=4592 Ack=1782 Win=12320 Len=0
20	3.234770	130.94.88.123	192.168.2.100	TCP	[TCP segment of a reassembled PDU]
21	3.235519	130.94.88.123	192.168.2.100	TLSv1	Application Data
22	3.235588	192.168.2.100	130.94.88.123	TCP	lv-jc > https [ACK] Seq=1782 Ack=6110 Win=128480 Len=0
23	3.737122	192.168.2.100	130.94.88.123	TCP	[TCP segment of a reassembled PDU]
24	3.737295	192.168.2.100	130.94.88.123	TLSv1	Application Data
25	4.151556	130.94.88.123	192.168.2.100	TCP	https > lv-jc [ACK] Seq=6110 Ack=3261 Win=15024 Len=0
26	4.151984	130.94.88.123	192.168.2.100	TCP	[TCP segment of a reassembled PDU]
27	4.152276	130.94.88.123	192.168.2.100	TLSv1	Application Data
28	4.152370	192.168.2.100	130.94.88.123	TCP	lv-jc > https [ACK] Seq=3261 Ack=7776 Win=128480 Len=0
29	7.996331	192.168.2.100	212.150.49.10	DNS	Standard query A mail.barak.net.il
30	8.025917	212.150.49.10	192.168.2.100	DNS	Standard query response A 194.90.6.40
31	8.077161	192.168.2.100	194.90.6.40	TCP	dynamic3d > pop3 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=1 TSV=0 TS
32	8.098732	194.90.6.40	192.168.2.100	TCP	pop3 > dynamic3d [SYN, ACK] Seq=0 Ack=1 Win=49580 Len=0 TSV=8290107
33	8.098776	192.168.2.100	194.90.6.40	TCP	dynamic3d > pop3 [ACK] Seq=1 Ack=1 Win=128480 Len=0 TSV=7813 TSER=8
34	8.118204	194.90.6.40	192.168.2.100	POP	S: +OK POP3 service
35	8.118745	192.168.2.100	194.90.6.40	POP	C: USER yoram-ndi.co.il
36	8.138633	194.90.6.40	192.168.2.100	TCP	pop3 > dynamic3d [ACK] Seq=19 Ack=23 Win=49580 Len=0 TSV=829010732
37	8.140050	194.90.6.40	192.168.2.100	POP	S: +OK password required for user yoram-ndi.co.il

Basic usage of Grep

Command-line text-search program in Linux

Some useful usage:

- Grep ‘word’ filename # find lines with ‘word’
- Grep –v ‘word’ filename # find lines without ‘word’
- Grep ‘^word’ filename # find lines beginning with ‘word’
- Grep ‘word’ filename > file2 # output lines with ‘word’ to file2
- ls -l | grep rwxrwxrwx # list files that have ‘rwxrwxrwx’ feature
- grep '^[0-4]‘ filename # find lines beginning with any of the numbers from 0-4
- Grep –c ‘word’ filename # find lines with ‘word’ and print out the number of these lines
- Grep –i ‘word’ filename # find lines with ‘word’ regardless of case

Many tutorials on grep online

- http://www.cyberciti.biz/faq/howto-use-grep-command-in-linux-unix/
- http://www.thegeekstuff.com/2009/03/15-practical-unix-grep-command-examples/

References

Wireshark Website

- http://www.wireshark.org

Wireshark Documentation

- http://www.wireshark.org/docs/

Wireshark Wiki

- http://wiki.wireshark.org

Network analysis Using Wireshark Cookbook

- http://www.amazon.com/Network-Analysis-Using-Wireshark-Cookbook/dp/1849517649

Q&A